

paper

Arezoo

0. Setup

```
library(AER)
library(BayesFactor)
library(BFpack)
library(brms)
library(broom)
library(dagitty)
library(devtools)
library(dplyr)
library(easystats)
library(emmeans)
library(GGally)
library(ggdag)
library(ggdist)
library(ggExtra)
library(ggplot2)
library(ggrepel)
library(GPArotation)
library(gridExtra)
library(haven)
library(infer)
library(janitor)
library(jpeg)
library(kableExtra)
library(knitr)
library(lavaan)
library(lme4)
library(logspline)
library(magrittr)
```

```
library(margins)
library(modelr)
library(patchwork)
library(pbkrtest)
library(psych)
library(purrr)
library(pwr)
library(readr)
library(readxl)
library(rstanarm)
library(scales)
library(semPlot)
library(stringr)
library(tidyLPA)
library(tidyr)
library(tidyverse)
```

Data

```
# adding data set
data <- read.csv("data.csv")

data$sex <- as.factor(data$sex)
data$ethnicity <- as.factor(data$ethnicity)
data$year <- as.factor(data$year)

str(data)
```

```
'data.frame':  390 obs. of  376 variables:
 $ N                : int  1 2 3 4 5 8 9 10 12 13 ...
 $ Progress          : int  100 100 100 100 100 100 100 100 100 100 ...
 $ Duration..in.seconds.: int  1415 3058 1550 1895 1068 1457 1001 43652 1350 11984 ...
 $ LS1               : chr  "QPPQ" "DDDE" "QPPQ" "QPPQ" ...
 $ LS2               : chr  "PRST" "BCDE" "PRST" "PRST" ...
 $ LS3               : chr  "PWXQ" "JVZK" "PWXQ" "PWXQ" ...
 $ LS4               : chr  "BCEF" "CDFG" "STWX" "STWX" ...
 $ LS5               : chr  "WXXW" "LMML" "GFFG" "GFFG" ...
 $ LS6               : chr  "CCRR" "AAPP" "QQBB" "QQBB" ...
 $ LS7               : chr  "EGFH" "ABDC" "IJLK" "EGFH" ...
```

\$ LS8	: chr	"BODQ"	"CERT"	"FHXZ"	"BODQ"	...
\$ LS9	: chr	"SEFT"	"SEFT"	"COPD"	"FUZG"	...
\$ LS10	: chr	"JCVC"	"CGCS"	"CGCS"	"CLXC"	...
\$ LS11	: chr	"VEGV"	"VEGV"	"XDBK"	"XDBK"	...
\$ LS12	: chr	"CEIZ"	"CEIZ"	"CGVZ"	"CGVZ"	...
\$ LS13	: chr	"VEBT"	"VEBT"	"VEBT"	"VEBT"	...
\$ LS14	: chr	"EJFK"	"PSQT"	"EJFK"	"GKHM"	...
\$ LS15	: chr	"LHEC"	"HJMQ"	"HJMQ"	"HJMQ"	...
\$ FC1_1	: chr	"5,3,2"	"8,7,6,4,1"	"3,5,7"	"3,5,2"	...
\$ FC1_2	: chr	"8,7,6,4,1"	"5,3,2"	"6,2,1"	"8,7,6,4,1"	...
\$ FC2_1	: chr	"6,4,2"	"7,4,2,3,6"	"2,3,1"	"6,4,2"	...
\$ FC2_2	: chr	"8,7,5,3,1"	"8,5,1"	"6,5,4"	"8,7,5,3,1"	...
\$ FC3_1	: chr	"7,4,1"	"6,4,3,2"	"3,2,1"	"7,4,1"	...
\$ FC3_2	: chr	"8,6,5,3,2"	"7,8,5,1"	"5,7,4"	"8,6,5,3,2"	...
\$ FC4_1	: chr	"8,6,3,2"	"7,5,4,1"	"4,2,3"	"8,6,3,2"	...
\$ FC4_2	: chr	"7,5,4,1"	"6,3,8,2"	"8,7,1"	"7,5,4,1"	...
\$ FC5_1	: chr	"6,8,3,1"	"8,7,4,5,3,2"	"2,6,4"	"8,5,4,2"	...
\$ FC5_2	: chr	"5,7,4,2"	"6,1"	"7,5,1"	"7,6,3,1"	...
\$ FC6_1	: chr	"7,6,4,1"	"2,5,4,1"	"3,1,4"	"6,4,8,5,3,2,1"	...
\$ FC6_2	: chr	"8,5,3,2"	"8,7,6,3"	"7,6,2"	"7"	...
\$ FC7_1	: chr	"8,6,3,1"	"8,6,5,3,1"	"2,8,1"	"7,5,3,8,2"	...
\$ FC7_2	: chr	"7,5,4,2"	"7,4,2"	"4,6,5"	"6,4,1"	...
\$ FC8_1	: chr	"6,5,3,1"	"8,5,2,4"	"5,3,1"	"6,5,4,1"	...
\$ FC8_2	: chr	"8,7,4,2"	"7,6,3,1"	"8,6,7"	"8,7,3,2"	...
\$ FC9_1	: chr	"2,1"	"6,4,1"	"5,1,2"	"6,2"	...
\$ FC9_2	: chr	"4,3,8"	"7,3,2"	"7,6,4"	"5,4,3"	...
\$ FC9_3	: chr	"5,7,6"	"8,5"	"3,8"	"8,7,1"	...
\$ FC10_1	: chr	"8,3,1"	"7,8,3,1"	"3,5,1"	"3,1"	...
\$ FC10_2	: chr	"5,6,2"	"5,4"	"7,6,4"	"7,4"	...
\$ FC10_3	: chr	"7,4"	"2,6"	"8,2"	"8,5,6,2"	...
\$ FC11_1	: chr	"8,1,3"	"6,4"	NA	"4,6"	...
\$ FC11_2	: chr	"6,4,2"	"3,1,5"	NA	"5,1,3"	...
\$ FC11_3	: chr	"7,5"	"2,8,7"	NA	"2,7,8"	...
\$ FC12_1	: chr	"1,5,6"	"4,1,2"	NA	"2,5,7"	...
\$ FC12_2	: chr	"2,8,4"	"3,6,8"	NA	"1,4,3"	...
\$ FC12_3	: chr	"3,7"	"5,7"	NA	"8,6"	...
\$ FC13_1	: chr	"2,5,1"	"1,4,6"	NA	"1,8,7"	...
\$ FC13_2	: chr	"3,6,4"	"2,8"	NA	"2,6"	...
\$ FC13_3	: chr	"7,8"	"3,7,5"	NA	"3,4,5"	...
\$ FC14_1	: chr	"1,3,6,8"	"5,2,3"	NA	"2,5"	...
\$ FC14_2	: chr	"2,5,4"	"7,8"	NA	"3,7,8"	...
\$ FC14_3	: chr	"7"	"1,4,6"	NA	"1,6,4"	...
\$ CN_1_1	: int	3	5	4	5	6 6 6 6 5 ...

\$ CN_1_2	: int	5 5 5 4 6 5 6 6 7 5 ...
\$ CN_1_3	: int	5 5 5 5 6 6 5 5 7 5 ...
\$ CN_1_4	: int	4 5 5 3 6 6 1 6 7 5 ...
\$ CN_1_5	: int	4 6 5 6 6 6 5 6 7 5 ...
\$ CN_1_6	: int	3 5 5 6 6 6 5 5 6 5 ...
\$ CN_1_7	: int	5 5 5 5 6 4 4 7 6 5 ...
\$ CN_1_8	: int	3 5 5 5 6 6 5 5 6 5 ...
\$ CN_2_1	: int	5 6 4 4 6 5 6 5 4 6 ...
\$ CN_2_2	: int	5 5 5 5 6 5 5 6 6 6 ...
\$ CN_2_3	: int	5 5 5 4 6 4 6 7 6 5 ...
\$ CN_2_4	: int	4 5 5 4 6 6 5 7 5 5 ...
\$ CN_2_5	: int	5 5 5 4 6 6 5 7 6 6 ...
\$ CN_2_6	: int	3 5 5 5 6 6 6 7 6 6 ...
\$ CN_2_7	: int	3 5 5 5 5 5 5 7 6 5 ...
\$ CN_2_8	: int	5 5 5 5 6 4 5 7 6 5 ...
\$ CN_3_1	: int	4 5 5 5 6 5 6 3 5 6 ...
\$ CN_3_2	: int	4 5 5 6 6 6 6 4 6 4 ...
\$ CN_3_3	: int	4 5 5 6 6 5 5 5 5 5 ...
\$ CN_3_4	: int	5 4 5 5 6 4 5 4 6 4 ...
\$ CN_3_5	: int	6 7 5 7 7 4 5 7 5 4 ...
\$ CN_3_6	: int	5 6 5 7 7 5 6 7 5 4 ...
\$ CN_3_7	: int	3 6 5 5 7 4 7 4 5 6 ...
\$ CN_3_8	: int	3 6 6 7 7 5 6 6 5 5 ...
\$ CN_4_1	: int	7 7 5 6 6 5 5 6 5 5 ...
\$ CN_4_2	: int	7 7 4 6 7 5 6 5 5 4 ...
\$ CN_4_3	: int	6 7 4 7 6 5 5 5 5 4 ...
\$ CN_4_4	: int	7 7 5 7 7 5 5 6 6 5 ...
\$ XXX1	: int	3 3 3 3 3 3 3 3 3 3 ...
\$ CN_4_5	: int	7 7 5 6 7 5 5 6 6 4 ...
\$ CN_4_6	: int	6 7 4 6 7 5 5 6 5 5 ...
\$ CN_4_7	: int	7 5 5 6 6 5 5 7 6 5 ...
\$ CN_4_8	: int	7 7 5 6 7 5 5 7 5 6 ...
\$ CN_5_1	: int	4 7 4 7 7 4 6 6 7 4 ...
\$ CN_5_2	: int	3 6 4 5 7 4 6 7 5 4 ...
\$ CN_5_3	: int	3 7 5 7 7 5 6 5 5 5 ...
\$ CN_5_4	: int	3 7 4 6 6 4 7 5 6 4 ...
\$ CN_5_5	: int	4 7 4 7 7 4 5 7 5 5 ...
\$ CN_5_6	: int	3 7 5 6 7 5 5 5 5 3 ...
\$ CN_5_7	: int	5 7 5 7 7 4 7 6 5 5 ...
\$ CN_5_8	: int	4 7 5 5 7 5 6 7 5 4 ...
\$ MA_1_1	: int	7 7 6 6 7 6 5 4 7 5 ...
\$ MA_1_2	: int	6 6 6 7 7 6 5 5 6 5 ...
\$ MA_1_3	: int	5 7 6 4 7 6 4 5 6 6 ...

```
$ MA_1_4          : int  4 7 7 7 7 6 5 7 6 6 ...
$ MA_1_5          : int  4 4 4 3 7 5 3 3 6 3 ...
$ MA_1_6          : int  7 7 6 6 7 7 5 6 7 6 ...
[list output truncated]
```

Descrip

```
age_summary<-data|>
  summarize(Variable= "Age", Mean=mean(age), Median=median(age),sd= sd(age), n= n())
```

```
descriptives <- describe(data[, c(
  "age", "gpa", "act", "sat_reading", "sat_math",
  "LS", "FC", "EQ",
  "CN_S", "CN_A", "CN_I", "CN_C", "CN_M",
  "M_S", "M_B", "M_Se", "M_Sa",
  "EI_Ch", "EI_Cu", "EI_I", "EI_E", "EI_P", "EI_D",
  "MC_S", "MC_P", "MC_A",
  "NE_N", "NE_O", "NE_C",
  "AR_1_1", "AR_1_2", "AR_1_3", "AR_1_4", "AR_1_5",
  "AR_2", "AR_3",
  "AI_1", "AI_2", "AI_3", "AI_4", "AI_5"
)])
```

```
descriptives <- descriptives[, c("n", "mean", "median", "sd", "min", "max")]
```

```
descriptives |> kable (caption= "Summary Statistics",digits=2)
```

Table 1: Summary Statistics

	n	mean	median	sd	min	max
age	390	19.93	20.00	1.44	18	30.00
gpa	354	3.65	3.73	0.37	2	4.25
act	125	32.88	34.00	3.39	13	36.00
sat_reading	230	727.08	740.00	57.03	370	800.00
sat_math	230	750.41	770.00	60.47	420	800.00
LS	386	9.93	11.00	3.47	1	15.00
FC	389	6.16	6.00	2.93	0	14.00
EQ	390	169.27	170.00	20.89	112	228.00
CN_S	390	44.47	44.00	6.56	22	56.00

	n	mean	median	sd	min	max
CN_A	390	41.57	42.00	7.88	19	56.00
CN_I	390	41.60	41.00	6.99	18	56.00
CN_C	390	45.67	46.00	6.39	30	56.00
CN_M	390	41.38	42.00	9.46	11	56.00
M_S	390	51.68	52.00	8.88	23	70.00
M_B	390	59.54	61.00	8.57	29	70.00
M_Se	390	57.71	58.00	8.46	32	70.00
M_Sa	390	52.28	52.00	8.06	26	70.00
EI_Ch	390	28.88	29.00	6.74	6	42.00
EI_Cu	390	28.79	29.00	6.43	10	42.00
EI_I	390	24.95	25.00	5.39	6	35.00
EI_E	390	20.18	21.00	5.82	5	35.00
EI_P	390	13.85	14.50	3.79	3	21.00
EI_D	390	20.46	21.00	6.11	5	35.00
MC_S	390	22.81	23.00	3.21	10	28.00
MC_P	390	13.02	13.00	3.44	3	21.00
MC_A	390	15.67	16.00	2.96	5	21.00
NE_N	390	47.15	47.00	11.04	16	77.00
NE_O	390	58.88	59.00	9.24	28	84.00
NE_C	390	60.38	61.00	8.97	29	82.00
AR_1_1	390	5.45	6.00	1.31	1	7.00
AR_1_2	390	4.48	5.00	1.54	1	7.00
AR_1_3	390	4.87	5.00	1.44	1	7.00
AR_1_4	390	5.41	6.00	1.20	1	7.00
AR_1_5	390	5.06	5.00	1.42	1	7.00
AR_2	390	5.26	5.00	1.25	1	7.00
AR_3	364	5.00	5.00	1.30	1	7.00
AI_1	390	5.62	6.00	1.27	1	7.00
AI_2	389	5.14	5.00	1.49	1	7.00
AI_3	390	5.82	6.00	1.19	1	7.00
AI_4	390	5.99	6.00	1.02	1	7.00
AI_5	390	5.93	6.00	1.30	1	7.00

```

categorical_summary <- data |>
  select(sex, ethnicity, year) |>
  pivot_longer(everything(),
    names_to = "Variable",
    values_to = "Category") |>
  filter(!is.na(Category)) |>
  count(Variable, Category) |>

```

```
group_by(Variable) |>
mutate(Percentage = round(n / sum(n) * 100, 1))

categorical_summary |> kable (caption= "Summary Statistics",digits=2)
```

Table 2: Summary Statistics

Variable	Category	n	Percentage
ethnicity	Asian	108	27.8
ethnicity	Asian American	31	8.0
ethnicity	Black or African American	26	6.7
ethnicity	Hispanic or Latino	29	7.5
ethnicity	prefer not to answer	11	2.8
ethnicity	two or more races	45	11.6
ethnicity	White or Caucasian	139	35.7
sex	Female	273	70.0
sex	Male	113	29.0
sex	other	2	0.5
sex	Prefer not to say	2	0.5
year	Fifth year or beyond	4	1.0
year	First year (Freshman)	57	14.7
year	Fourth year (Senior)	99	25.4
year	Second year (Sophomore)	134	34.4
year	Third year (Junior)	95	24.4

Reliability

```
#psych::alpha(data[, paste0("EQ", 1:33)]) |> print ()
```

```
# Emotional Intelligence
```

```
psych::alpha(data[, paste0("EQ", 1:33)])$total$raw_alpha
```

```
[1] 0.8986268
```

```
# Core Needs
```

```
psych::alpha(data[, paste0("CN_1_", 1:8)])$total$raw_alpha # Security
```

```
[1] 0.8199992
```

```
psych::alpha(data[, paste0("CN_2_", 1:8)])$total$raw_alpha # Acceptance
```

```
[1] 0.8866242
```

```
psych::alpha(data[, paste0("CN_3_", 1:8)])$total$raw_alpha # Initiative
```

```
[1] 0.8422906
```

```
psych::alpha(data[, paste0("CN_4_", 1:8)])$total$raw_alpha # Competence
```

```
[1] 0.8940293
```

```
psych::alpha(data[, paste0("CN_5_", 1:8)])$total$raw_alpha # Meaning
```

```
[1] 0.9216608
```

```
# Maslow
```

```
psych::alpha(data[, paste0("MA_1_", 1:10)])$total$raw_alpha # Safety
```

```
[1] 0.8634134
```

```
psych::alpha(data[, paste0("MA_2_", 1:10)])$total$raw_alpha # Belonging
```

```
[1] 0.8818991
```

```
psych::alpha(data[, paste0("MA_3_", 1:10)])$total$raw_alpha # Self-esteem
```

```
[1] 0.9214845
```

```
psych::alpha(data[, paste0("MA_4_", 1:10)])$total$raw_alpha # Self-actualization
```

```
[1] 0.8233781
```

```
# Extrinsic / Intrinsic Motives
```

```
psych::alpha(data[, paste0("EI_1_", 1:6)])$total$raw_alpha # Challenge
```

```
[1] 0.8892558
```

```
psych::alpha(data[, paste0("EI_2_", 1:6)])$total$raw_alpha # Curiosity
```

```
[1] 0.8173063
```

```
psych::alpha(data[, paste0("EI_3_", 1:5)])$total$raw_alpha # Independence Mastery
```

```
[1] 0.868526
```

```
psych::alpha(data[, paste0("EI_4_", 1:5)])$total$raw_alpha # Easy Work
```

```
[1] 0.8323405
```

```
psych::alpha(data[, paste0("EI_5_", 1:3)])$total$raw_alpha # Pleasing Teacher
```

```
[1] 0.7585386
```

```
psych::alpha(data[, paste0("EI_6_", 1:5)])$total$raw_alpha # Dependence on Teacher
```

```
[1] 0.8274458
```

```
# McClelland
```

```
psych::alpha(data[, paste0("MC_1_", 1:4)])$total$raw_alpha # Self-achievement
```

```
[1] 0.6759667
```

```
psych::alpha(data[, paste0("MC_2_", 1:3)])$total$raw_alpha # Power
```

```
[1] 0.6974886
```

```
psych::alpha(data[, paste0("MC_3_", 1:3)])$total$raw_alpha # Affiliation
```

```
[1] 0.4760393
```

```
# NEO
```

```
psych::alpha(data[, paste0("NE_1_", 1:12)])$total$raw_alpha # Neuroticism
```

```
[1] 0.830528
```

```
psych::alpha(data[, paste0("NE_2_", 1:12)])$total$raw_alpha # Openness
```

```
[1] 0.7051049
```

```
psych::alpha(data[, paste0("NE_3_", 1:12)])$total$raw_alpha # Conscientiousness
```

```
[1] 0.7995361
```

```
# LS and FC with KR-20
```

```
# Letter Sets
```

```
psych::alpha(data[, paste0("LS", 1:15, "_score")])$total$raw_alpha
```

Warning in psych::alpha(data[, paste0("LS", 1:15, "_score")]): Some items were negatively correlated with the first principal component and should be reversed.

To do this, run the function again with the 'check.keys=TRUE' option

Some items (LS15_score) were negatively correlated with the first principal component and probably should be reversed.

To do this, run the function again with the 'check.keys=TRUE' option

```
[1] 0.8183421
```

```
psych::alpha(data[, paste0("LS", 1:15, "_score")])$item.stats
```

Warning in psych::alpha(data[, paste0("LS", 1:15, "_score")]): Some items were negatively correlated with the first principal component and should be reversed.

To do this, run the function again with the 'check.keys=TRUE' option

Some items (LS15_score) were negatively correlated with the first principal component and probably should be reversed.

To do this, run the function again with the 'check.keys=TRUE' option

	n	raw.r	std.r	r.cor	r.drop	mean
LS1_score	386	0.55031749	0.58441579	0.5537765	0.4792727	0.8782383
LS2_score	386	0.57195716	0.59597121	0.5610893	0.4928572	0.8341969
LS3_score	386	0.58508257	0.62157699	0.5926120	0.5218880	0.8937824
LS4_score	386	0.56022839	0.56499444	0.5210536	0.4646261	0.7461140
LS5_score	386	0.63227502	0.65175819	0.6321893	0.5594917	0.8264249
LS6_score	386	0.58672472	0.59745573	0.5566885	0.5030987	0.8005181
LS7_score	386	0.54222530	0.54286213	0.4889477	0.4424842	0.7331606
LS8_score	386	0.57550849	0.55945269	0.5132882	0.4683309	0.5880829
LS9_score	386	0.61944579	0.60003653	0.5664357	0.5178794	0.4818653
LS10_score	386	0.67295925	0.66627051	0.6436852	0.5883559	0.6787565
LS11_score	386	0.63298029	0.61270548	0.5829123	0.5350700	0.5751295
LS12_score	386	0.59001145	0.58392197	0.5455269	0.4936973	0.7046632
LS13_score	386	0.43771979	0.41453026	0.3430749	0.3104761	0.5284974
LS14_score	386	0.48329082	0.45992451	0.3896164	0.3611877	0.4818653
LS15_score	386	-0.03741213	-0.03141938	-0.1595485	-0.1464866	0.1787565

	sd
LS1_score	0.3274347
LS2_score	0.3723865
LS3_score	0.3085158
LS4_score	0.4357980
LS5_score	0.3792353
LS6_score	0.4001295
LS7_score	0.4428818
LS8_score	0.4928190
LS9_score	0.5003195
LS10_score	0.4675601
LS11_score	0.4949649
LS12_score	0.4567861
LS13_score	0.4998351
LS14_score	0.5003195
LS15_score	0.3836455

```
# Figure Classification
psych::alpha(data[, paste0("FC", 1:14)])$total$raw_alpha
```

```
[1] 0.7955718
```

Correlation

```
# Load required package
# Select variables for correlations
cor_data <- data[, c(
  "CN_S", "CN_A", "CN_I", "CN_C", "CN_M",
  "M_S", "M_B", "M_Se", "M_Sa",
  "EI_Ch", "EI_Cu", "EI_I", "EI_E", "EI_P", "EI_D",
  "MC_S", "MC_P",
  "NE_N", "NE_O", "NE_C", "EQ",
  "LS", "FC", "gpa")]

variable_labels <- c(
  CN_S = "CN_Security",
  CN_A = "CN_Acceptance",
  CN_I = "CN_Initiative",
  CN_C = "CN_Compotence",
  CN_M = "CN_Meaning",
  M_S = "M_Safety",
  M_B = "M_Belonging",
  M_Se = "M_Self-esteem",
  M_Sa = "M_Self-actualization",
  EI_Ch = "EI_Challenge",
  EI_Cu = "EI_Curiosity",
  EI_I = "EI_Independence Mastery",
  EI_E = "EI_Easy work",
  EI_P = "EI_Pleasing",
  EI_D = "EI_Dependence on teacher",
  MC_S = "MC_Self-achievement",
  MC_P = "MC_Power",

  EQ = "Emotional Intelligence",
  NE_N = "NE_Neuroticism",
  NE_O = "NE_Openness",
  NE_C = "NE_Conscientiousness",
  LS = "Letter Sets",
  FC = "Figure Classification",
  NS_S = "Security satisfaction",
  NS_A = "Acceptance satisfaction",
  NS_I = "Initiative satisfaction",
```

```

  NS_C = "Self-efficacy satisfaction",
  NS_M = "Meaning satisfaction",
  gpa = "GPA"
)

names(cor_data) <- unname(variable_labels[names(cor_data)])

intercor_matrix <- cor(cor_data, use = "pairwise.complete.obs", method = "pearson")

print(intercor_matrix)

```

	CN_Security	CN_Acceptance	CN_Initiative	
CN_Security	1.0000000000	0.663484032	0.53300069	
CN_Acceptance	0.6634840315	1.0000000000	0.55038056	
CN_Initiative	0.5330006854	0.550380559	1.00000000	
CN_Competence	0.5768268480	0.488480266	0.69712508	
CN_Meaning	0.4821872957	0.467468495	0.61381630	
M_Safety	0.5090872217	0.321604258	0.35718742	
M_Belonging	0.5656698827	0.448376915	0.36027607	
M_Self-esteem	0.5932860068	0.461417046	0.58801900	
M_Self-actualization	0.3792224152	0.425917831	0.58816997	
EI_Challenge	0.0801164315	0.151013552	0.35226347	
EI_Curiosity	0.2206242515	0.262404321	0.40317234	
EI_Independence Mastery	0.1141094091	0.098354285	0.26279391	
EI_Easy work	0.0996994300	0.001456188	-0.10349605	
EI_Pleasing	0.0717706432	0.013437565	-0.11384500	
EI_Dependence on teacher	0.1976054689	0.183910357	0.08053243	
MC_Self-achievement	0.2651129756	0.233037992	0.41872634	
MC_Power	0.1086456450	0.076214871	0.18894545	
NE_Neuroticism	0.1109826662	0.099468536	-0.03793065	
NE_Openness	0.1102219831	0.175860934	0.10573859	
NE_Conscientiousness	0.2305387208	0.203111511	0.24726585	
Emotional Intelligence	0.2877458214	0.281607300	0.43373623	
Letter Sets	-0.0603682099	-0.083451613	-0.12436229	
Figure Classification	-0.0547436816	-0.051177969	-0.21344052	
GPA	0.0004787008	0.057566989	-0.09246857	
	CN_Competence	CN_Meaning	M_Safety	M_Belonging
CN_Security	0.5768268480	0.482187296	0.50908722	0.56566988
CN_Acceptance	0.4884802657	0.467468495	0.32160426	0.44837691
CN_Initiative	0.6971250820	0.613816296	0.35718742	0.36027607
CN_Competence	1.0000000000	0.597788058	0.44295511	0.41611965

CN_Meaning	0.5977880581	1.0000000000	0.41816707	0.41255781
M_Safety	0.4429551135	0.418167073	1.00000000	0.39903988
M_Belonging	0.4161196545	0.412557807	0.39903988	1.00000000
M_Self-esteem	0.7198862878	0.581344865	0.52708542	0.61405307
M_Self-actualization	0.4855572253	0.482808618	0.30641109	0.42277839
EI_Challenge	0.2692522559	0.322613268	0.03345958	0.14546967
EI_Curiosity	0.3209193826	0.416982531	0.15493423	0.25179307
EI_Independence Mastery	0.2414139815	0.231000417	0.15297408	0.07820184
EI_Easy work	-0.0421848180	-0.029239573	0.21671306	0.04683165
EI_Pleasing	0.0007547254	-0.042975025	0.15315228	0.05250666
EI_Dependence on teacher	0.1080213021	0.165443073	0.18508170	0.21699956
MC_Self-achievement	0.4529304801	0.375517618	0.19285961	0.38427611
MC_Power	0.2425648281	0.163451827	0.26062418	0.18664104
NE_Neuroticism	0.0055731489	0.003100202	0.18214791	0.01737406
NE_Openness	0.0495968040	0.078001782	-0.12262052	0.11087079
NE_Conscientiousness	0.2836082121	0.296113136	0.24447156	0.30323472
Emotional Intelligence	0.3793600460	0.389222624	0.14966472	0.41307600
Letter Sets	-0.0615562359	-0.173030856	-0.12267344	0.03065208
Figure Classification	-0.1137167377	-0.162965133	-0.15193046	-0.02276039
GPA	-0.0263324090	-0.072777284	-0.08130712	0.10233505
	M_Self-esteem	M_Self-actualization	EI_Challenge	
CN_Security	0.593286007	0.37922242	0.08011643	
CN_Acceptance	0.461417046	0.42591783	0.15101355	
CN_Initiative	0.588018996	0.58816997	0.35226347	
CN_Competence	0.719886288	0.48555723	0.26925226	
CN_Meaning	0.581344865	0.48280862	0.32261327	
M_Safety	0.527085423	0.30641109	0.03345958	
M_Belonging	0.614053070	0.42277839	0.14546967	
M_Self-esteem	1.000000000	0.61817064	0.23024485	
M_Self-actualization	0.618170642	1.00000000	0.27352684	
EI_Challenge	0.230244852	0.27352684	1.00000000	
EI_Curiosity	0.306148766	0.36352734	0.75882685	
EI_Independence Mastery	0.167364821	0.13329756	0.59276670	
EI_Easy work	0.033073260	-0.05740910	-0.69140791	
EI_Pleasing	0.044558892	-0.03398211	-0.45313859	
EI_Dependence on teacher	0.162796918	0.08897508	-0.14490814	
MC_Self-achievement	0.479859003	0.44546411	0.58659615	
MC_Power	0.303227304	0.18384150	0.06751288	
NE_Neuroticism	0.002586548	-0.12015388	-0.25419728	
NE_Openness	0.103350041	0.26101722	0.21760015	
NE_Conscientiousness	0.280445613	0.18682981	0.32539704	
Emotional Intelligence	0.456026741	0.49998446	0.44228244	
Letter Sets	-0.030881628	-0.11800527	0.01930366	

Figure Classification	-0.091865379	-0.13448426	0.03202353
GPA	-0.029200203	-0.17892702	0.11079319
	EI_Curiosity	EI_Independence	Mastery
CN_Security	0.22062425	0.11410941	0.099699430
CN_Acceptance	0.26240432	0.09835429	0.001456188
CN_Initiative	0.40317234	0.26279391	-0.103496054
CN_Competence	0.32091938	0.24141398	-0.042184818
CN_Meaning	0.41698253	0.23100042	-0.029239573
M_Safety	0.15493423	0.15297408	0.216713057
M_Belonging	0.25179307	0.07820184	0.046831652
M_Self-esteem	0.30614877	0.16736482	0.033073260
M_Self-actualization	0.36352734	0.13329756	-0.057409099
EI_Challenge	0.75882685	0.59276670	-0.691407911
EI_Curiosity	1.00000000	0.47309479	-0.507426395
EI_Independence	0.47309479	1.00000000	-0.405823855
EI_Easy work	-0.50742640	-0.40582386	1.000000000
EI_Pleasing	-0.43054442	-0.26478368	0.636395108
EI_Dependence on teacher	0.05674784	-0.30640263	0.290895693
MC_Self-achievement	0.49848352	0.37959811	-0.424420189
MC_Power	0.13497054	0.02914944	0.059892169
NE_Neuroticism	-0.21819675	-0.14142621	0.310802579
NE_Openness	0.23630225	0.05918322	-0.275266105
NE_Conscientiousness	0.30236775	0.26152715	-0.240660464
Emotional Intelligence	0.51134336	0.25570133	-0.263952479
Letter Sets	-0.11490995	0.06468436	-0.073125909
Figure Classification	-0.04457247	0.04283845	-0.146022765
GPA	0.07250894	0.09945813	-0.159062021
	EI_Pleasing	EI_Dependence on teacher	
CN_Security	0.0717706432	0.19760547	
CN_Acceptance	0.0134375652	0.18391036	
CN_Initiative	-0.1138449977	0.08053243	
CN_Competence	0.0007547254	0.10802130	
CN_Meaning	-0.0429750246	0.16544307	
M_Safety	0.1531522816	0.18508170	
M_Belonging	0.0525066597	0.21699956	
M_Self-esteem	0.0445588919	0.16279692	
M_Self-actualization	-0.0339821080	0.08897508	
EI_Challenge	-0.4531385858	-0.14490814	
EI_Curiosity	-0.4305444184	0.05674784	
EI_Independence	-0.2647836850	-0.30640263	
EI_Easy work	0.6363951076	0.29089569	
EI_Pleasing	1.0000000000	0.24750942	
EI_Dependence on teacher	0.2475094186	1.00000000	

MC_Self-achievement	-0.2249210451	-0.06933198
MC_Power	0.0209222352	0.15547506
NE_Neuroticism	0.2143169837	0.06738510
NE_Openness	-0.1953551754	-0.05374545
NE_Conscientiousness	-0.1497863568	0.14006457
Emotional Intelligence	-0.1553007019	0.07097608
Letter Sets	0.0131172968	-0.11734590
Figure Classification	-0.1062366626	-0.15483213
GPA	-0.1435770263	0.02682367
	MC_Self-achievement	MC_Power NE_Neuroticism
CN_Security	0.26511298	0.108645645 0.110982666
CN_Acceptance	0.23303799	0.076214871 0.099468536
CN_Initiative	0.41872634	0.188945449 -0.037930650
CN_Competence	0.45293048	0.242564828 0.005573149
CN_Meaning	0.37551762	0.163451827 0.003100202
M_Safety	0.19285961	0.260624175 0.182147915
M_Belonging	0.38427611	0.186641035 0.017374056
M_Self-esteem	0.47985900	0.303227304 0.002586548
M_Self-actualization	0.44546411	0.183841499 -0.120153881
EI_Challenge	0.58659615	0.067512875 -0.254197278
EI_Curiosity	0.49848352	0.134970538 -0.218196746
EI_Independence Mastery	0.37959811	0.029149436 -0.141426206
EI_Easy work	-0.42442019	0.059892169 0.310802579
EI_Pleasing	-0.22492105	0.020922235 0.214316984
EI_Dependence on teacher	-0.06933198	0.155475056 0.067385103
MC_Self-achievement	1.00000000	0.161899585 -0.154184612
MC_Power	0.16189959	1.000000000 0.044051668
NE_Neuroticism	-0.15418461	0.044051668 1.000000000
NE_Openness	0.20811755	0.008493264 0.042079358
NE_Conscientiousness	0.44595487	0.003326708 -0.321788622
Emotional Intelligence	0.50364356	0.176639099 -0.425319270
Letter Sets	0.07724483	-0.097694388 0.032933372
Figure Classification	-0.02666903	-0.113333367 0.052661754
GPA	0.08567961	0.065328635 -0.010905513
	NE_Openness	NE_Conscientiousness
CN_Security	0.110221983	0.230538721
CN_Acceptance	0.175860934	0.203111511
CN_Initiative	0.105738591	0.247265845
CN_Competence	0.049596804	0.283608212
CN_Meaning	0.078001782	0.296113136
M_Safety	-0.122620517	0.244471558
M_Belonging	0.110870788	0.303234716
M_Self-esteem	0.103350041	0.280445613

M_Self-actualization	0.261017218	0.186829814
EI_Challenge	0.217600146	0.325397041
EI_Curiosity	0.236302248	0.302367753
EI_Independence Mastery	0.059183217	0.261527149
EI_Easy work	-0.275266105	-0.240660464
EI_Pleasing	-0.195355175	-0.149786357
EI_Dependence on teacher	-0.053745455	0.140064574
MC_Self-achievement	0.208117548	0.445954868
MC_Power	0.008493264	0.003326708
NE_Neuroticism	0.042079358	-0.321788622
NE_Openness	1.000000000	-0.102340924
NE_Conscientiousness	-0.102340924	1.000000000
Emotional Intelligence	0.247773260	0.419333532
Letter Sets	0.062444116	0.009103993
Figure Classification	0.084112183	-0.063157691
GPA	0.006135441	0.205706091
	Emotional Intelligence	Letter Sets
CN_Security	0.28774582	-0.060368210
CN_Acceptance	0.28160730	-0.083451613
CN_Initiative	0.43373623	-0.124362289
CN_Competence	0.37936005	-0.061556236
CN_Meaning	0.38922262	-0.173030856
M_Safety	0.14966472	-0.122673444
M_Belonging	0.41307600	0.030652082
M_Self-esteem	0.45602674	-0.030881628
M_Self-actualization	0.49998446	-0.118005268
EI_Challenge	0.44228244	0.019303656
EI_Curiosity	0.51134336	-0.114909949
EI_Independence Mastery	0.25570133	0.064684360
EI_Easy work	-0.26395248	-0.073125909
EI_Pleasing	-0.15530070	0.013117297
EI_Dependence on teacher	0.07097608	-0.117345897
MC_Self-achievement	0.50364356	0.077244830
MC_Power	0.17663910	-0.097694388
NE_Neuroticism	-0.42531927	0.032933372
NE_Openness	0.24777326	0.062444116
NE_Conscientiousness	0.41933353	0.009103993
Emotional Intelligence	1.00000000	-0.112720570
Letter Sets	-0.11272057	1.000000000
Figure Classification	-0.12546414	0.456398124
GPA	0.07426018	0.063219463
	Figure Classification	GPA
CN_Security	-0.05474368	0.0004787008

CN_Acceptance	-0.05117797	0.0575669887
CN_Initiative	-0.21344052	-0.0924685729
CN_Competence	-0.11371674	-0.0263324090
CN_Meaning	-0.16296513	-0.0727772843
M_Safety	-0.15193046	-0.0813071199
M_Belonging	-0.02276039	0.1023350503
M_Self-esteem	-0.09186538	-0.0292002029
M_Self-actualization	-0.13448426	-0.1789270226
EI_Challenge	0.03202353	0.1107931879
EI_Curiosity	-0.04457247	0.0725089358
EI_Independence Mastery	0.04283845	0.0994581347
EI_Easy work	-0.14602276	-0.1590620210
EI_Pleasing	-0.10623666	-0.1435770263
EI_Dependence on teacher	-0.15483213	0.0268236691
MC_Self-achievement	-0.02666903	0.0856796139
MC_Power	-0.11333337	0.0653286351
NE_Neuroticism	0.05266175	-0.0109055134
NE_Openness	0.08411218	0.0061354410
NE_Conscientiousness	-0.06315769	0.2057060914
Emotional Intelligence	-0.12546414	0.0742601785
Letter Sets	0.45639812	0.0632194628
Figure Classification	1.00000000	0.1114047545
GPA	0.11140475	1.0000000000

```
corr_test <- psych::corr.test(cor_data, use = "pairwise", method = "pearson",
                             adjust = "none")
print(corr_test$r)|> kable (digits=2) # Correlation coefficients
```

	CN_Security	CN_Acceptance	CN_Initiative
CN_Security	1.0000000000	0.663484032	0.53300069
CN_Acceptance	0.6634840315	1.0000000000	0.55038056
CN_Initiative	0.5330006854	0.550380559	1.00000000
CN_Competence	0.5768268480	0.488480266	0.69712508
CN_Meaning	0.4821872957	0.467468495	0.61381630
M_Safety	0.5090872217	0.321604258	0.35718742
M_Belonging	0.5656698827	0.448376915	0.36027607
M_Self-esteem	0.5932860068	0.461417046	0.58801900
M_Self-actualization	0.3792224152	0.425917831	0.58816997
EI_Challenge	0.0801164315	0.151013552	0.35226347
EI_Curiosity	0.2206242515	0.262404321	0.40317234
EI_Independence Mastery	0.1141094091	0.098354285	0.26279391
EI_Easy work	0.0996994300	0.001456188	-0.10349605

EI_Pleasing	0.0717706432	0.013437565	-0.11384500	
EI_Dependence on teacher	0.1976054689	0.183910357	0.08053243	
MC_Self-achievement	0.2651129756	0.233037992	0.41872634	
MC_Power	0.1086456450	0.076214871	0.18894545	
NE_Neuroticism	0.1109826662	0.099468536	-0.03793065	
NE_Openness	0.1102219831	0.175860934	0.10573859	
NE_Conscientiousness	0.2305387208	0.203111511	0.24726585	
Emotional Intelligence	0.2877458214	0.281607300	0.43373623	
Letter Sets	-0.0603682099	-0.083451613	-0.12436229	
Figure Classification	-0.0547436816	-0.051177969	-0.21344052	
GPA	0.0004787008	0.057566989	-0.09246857	
	CN_Competence	CN_Meaning	M_Safety	M_Belonging
CN_Security	0.5768268480	0.482187296	0.50908722	0.56566988
CN_Acceptance	0.4884802657	0.467468495	0.32160426	0.44837691
CN_Initiative	0.6971250820	0.613816296	0.35718742	0.36027607
CN_Competence	1.0000000000	0.597788058	0.44295511	0.41611965
CN_Meaning	0.5977880581	1.0000000000	0.41816707	0.41255781
M_Safety	0.4429551135	0.418167073	1.00000000	0.39903988
M_Belonging	0.4161196545	0.412557807	0.39903988	1.00000000
M_Self-esteem	0.7198862878	0.581344865	0.52708542	0.61405307
M_Self-actualization	0.4855572253	0.482808618	0.30641109	0.42277839
EI_Challenge	0.2692522559	0.322613268	0.03345958	0.14546967
EI_Curiosity	0.3209193826	0.416982531	0.15493423	0.25179307
EI_Independence Mastery	0.2414139815	0.231000417	0.15297408	0.07820184
EI_Easy work	-0.0421848180	-0.029239573	0.21671306	0.04683165
EI_Pleasing	0.0007547254	-0.042975025	0.15315228	0.05250666
EI_Dependence on teacher	0.1080213021	0.165443073	0.18508170	0.21699956
MC_Self-achievement	0.4529304801	0.375517618	0.19285961	0.38427611
MC_Power	0.2425648281	0.163451827	0.26062418	0.18664104
NE_Neuroticism	0.0055731489	0.003100202	0.18214791	0.01737406
NE_Openness	0.0495968040	0.078001782	-0.12262052	0.11087079
NE_Conscientiousness	0.2836082121	0.296113136	0.24447156	0.30323472
Emotional Intelligence	0.3793600460	0.389222624	0.14966472	0.41307600
Letter Sets	-0.0615562359	-0.173030856	-0.12267344	0.03065208
Figure Classification	-0.1137167377	-0.162965133	-0.15193046	-0.02276039
GPA	-0.0263324090	-0.072777284	-0.08130712	0.10233505
	M_Self-esteem	M_Self-actualization	EI_Challenge	
CN_Security	0.593286007	0.37922242	0.08011643	
CN_Acceptance	0.461417046	0.42591783	0.15101355	
CN_Initiative	0.588018996	0.58816997	0.35226347	
CN_Competence	0.719886288	0.48555723	0.26925226	
CN_Meaning	0.581344865	0.48280862	0.32261327	
M_Safety	0.527085423	0.30641109	0.03345958	

M_Belonging	0.614053070	0.42277839	0.14546967
M_Self-esteem	1.000000000	0.61817064	0.23024485
M_Self-actualization	0.618170642	1.00000000	0.27352684
EI_Challenge	0.230244852	0.27352684	1.00000000
EI_Curiosity	0.306148766	0.36352734	0.75882685
EI_Independence Mastery	0.167364821	0.13329756	0.59276670
EI_Easy work	0.033073260	-0.05740910	-0.69140791
EI_Pleasing	0.044558892	-0.03398211	-0.45313859
EI_Dependence on teacher	0.162796918	0.08897508	-0.14490814
MC_Self-achievement	0.479859003	0.44546411	0.58659615
MC_Power	0.303227304	0.18384150	0.06751288
NE_Neuroticism	0.002586548	-0.12015388	-0.25419728
NE_Openness	0.103350041	0.26101722	0.21760015
NE_Conscientiousness	0.280445613	0.18682981	0.32539704
Emotional Intelligence	0.456026741	0.49998446	0.44228244
Letter Sets	-0.030881628	-0.11800527	0.01930366
Figure Classification	-0.091865379	-0.13448426	0.03202353
GPA	-0.029200203	-0.17892702	0.11079319
	EI_Curiosity	EI_Independence Mastery	EI_Easy work
CN_Security	0.22062425	0.11410941	0.099699430
CN_Acceptance	0.26240432	0.09835429	0.001456188
CN_Initiative	0.40317234	0.26279391	-0.103496054
CN_Competence	0.32091938	0.24141398	-0.042184818
CN_Meaning	0.41698253	0.23100042	-0.029239573
M_Safety	0.15493423	0.15297408	0.216713057
M_Belonging	0.25179307	0.07820184	0.046831652
M_Self-esteem	0.30614877	0.16736482	0.033073260
M_Self-actualization	0.36352734	0.13329756	-0.057409099
EI_Challenge	0.75882685	0.59276670	-0.691407911
EI_Curiosity	1.00000000	0.47309479	-0.507426395
EI_Independence Mastery	0.47309479	1.00000000	-0.405823855
EI_Easy work	-0.50742640	-0.40582386	1.000000000
EI_Pleasing	-0.43054442	-0.26478368	0.636395108
EI_Dependence on teacher	0.05674784	-0.30640263	0.290895693
MC_Self-achievement	0.49848352	0.37959811	-0.424420189
MC_Power	0.13497054	0.02914944	0.059892169
NE_Neuroticism	-0.21819675	-0.14142621	0.310802579
NE_Openness	0.23630225	0.05918322	-0.275266105
NE_Conscientiousness	0.30236775	0.26152715	-0.240660464
Emotional Intelligence	0.51134336	0.25570133	-0.263952479
Letter Sets	-0.11490995	0.06468436	-0.073125909
Figure Classification	-0.04457247	0.04283845	-0.146022765
GPA	0.07250894	0.09945813	-0.159062021

	EI_Pleasing	EI_Dependence on teacher	
CN_Security	0.0717706432	0.19760547	
CN_Acceptance	0.0134375652	0.18391036	
CN_Initiative	-0.1138449977	0.08053243	
CN_Competence	0.0007547254	0.10802130	
CN_Meaning	-0.0429750246	0.16544307	
M_Safety	0.1531522816	0.18508170	
M_Belonging	0.0525066597	0.21699956	
M_Self-esteem	0.0445588919	0.16279692	
M_Self-actualization	-0.0339821080	0.08897508	
EI_Challenge	-0.4531385858	-0.14490814	
EI_Curiosity	-0.4305444184	0.05674784	
EI_Independence Mastery	-0.2647836850	-0.30640263	
EI_Easy work	0.6363951076	0.29089569	
EI_Pleasing	1.0000000000	0.24750942	
EI_Dependence on teacher	0.2475094186	1.00000000	
MC_Self-achievement	-0.2249210451	-0.06933198	
MC_Power	0.0209222352	0.15547506	
NE_Neuroticism	0.2143169837	0.06738510	
NE_Openness	-0.1953551754	-0.05374545	
NE_Conscientiousness	-0.1497863568	0.14006457	
Emotional Intelligence	-0.1553007019	0.07097608	
Letter Sets	0.0131172968	-0.11734590	
Figure Classification	-0.1062366626	-0.15483213	
GPA	-0.1435770263	0.02682367	
	MC_Self-achievement	MC_Power	NE_Neuroticism
CN_Security	0.26511298	0.108645645	0.110982666
CN_Acceptance	0.23303799	0.076214871	0.099468536
CN_Initiative	0.41872634	0.188945449	-0.037930650
CN_Competence	0.45293048	0.242564828	0.005573149
CN_Meaning	0.37551762	0.163451827	0.003100202
M_Safety	0.19285961	0.260624175	0.182147915
M_Belonging	0.38427611	0.186641035	0.017374056
M_Self-esteem	0.47985900	0.303227304	0.002586548
M_Self-actualization	0.44546411	0.183841499	-0.120153881
EI_Challenge	0.58659615	0.067512875	-0.254197278
EI_Curiosity	0.49848352	0.134970538	-0.218196746
EI_Independence Mastery	0.37959811	0.029149436	-0.141426206
EI_Easy work	-0.42442019	0.059892169	0.310802579
EI_Pleasing	-0.22492105	0.020922235	0.214316984
EI_Dependence on teacher	-0.06933198	0.155475056	0.067385103
MC_Self-achievement	1.00000000	0.161899585	-0.154184612
MC_Power	0.16189959	1.000000000	0.044051668

NE_Neuroticism	-0.15418461	0.044051668	1.000000000
NE_Openness	0.20811755	0.008493264	0.042079358
NE_Conscientiousness	0.44595487	0.003326708	-0.321788622
Emotional Intelligence	0.50364356	0.176639099	-0.425319270
Letter Sets	0.07724483	-0.097694388	0.032933372
Figure Classification	-0.02666903	-0.113333367	0.052661754
GPA	0.08567961	0.065328635	-0.010905513
	NE_Openness	NE_Conscientiousness	
CN_Security	0.110221983	0.230538721	
CN_Acceptance	0.175860934	0.203111511	
CN_Initiative	0.105738591	0.247265845	
CN_Competence	0.049596804	0.283608212	
CN_Meaning	0.078001782	0.296113136	
M_Safety	-0.122620517	0.244471558	
M_Belonging	0.110870788	0.303234716	
M_Self-esteem	0.103350041	0.280445613	
M_Self-actualization	0.261017218	0.186829814	
EI_Challenge	0.217600146	0.325397041	
EI_Curiosity	0.236302248	0.302367753	
EI_Independence Mastery	0.059183217	0.261527149	
EI_Easy work	-0.275266105	-0.240660464	
EI_Pleasing	-0.195355175	-0.149786357	
EI_Dependence on teacher	-0.053745455	0.140064574	
MC_Self-achievement	0.208117548	0.445954868	
MC_Power	0.008493264	0.003326708	
NE_Neuroticism	0.042079358	-0.321788622	
NE_Openness	1.000000000	-0.102340924	
NE_Conscientiousness	-0.102340924	1.000000000	
Emotional Intelligence	0.247773260	0.419333532	
Letter Sets	0.062444116	0.009103993	
Figure Classification	0.084112183	-0.063157691	
GPA	0.006135441	0.205706091	
	Emotional Intelligence	Letter Sets	
CN_Security	0.28774582	-0.060368210	
CN_Acceptance	0.28160730	-0.083451613	
CN_Initiative	0.43373623	-0.124362289	
CN_Competence	0.37936005	-0.061556236	
CN_Meaning	0.38922262	-0.173030856	
M_Safety	0.14966472	-0.122673444	
M_Belonging	0.41307600	0.030652082	
M_Self-esteem	0.45602674	-0.030881628	
M_Self-actualization	0.49998446	-0.118005268	
EI_Challenge	0.44228244	0.019303656	

EI_Curiosity	0.51134336	-0.114909949
EI_Independence Mastery	0.25570133	0.064684360
EI_Easy work	-0.26395248	-0.073125909
EI_Pleasing	-0.15530070	0.013117297
EI_Dependence on teacher	0.07097608	-0.117345897
MC_Self-achievement	0.50364356	0.077244830
MC_Power	0.17663910	-0.097694388
NE_Neuroticism	-0.42531927	0.032933372
NE_Openness	0.24777326	0.062444116
NE_Conscientiousness	0.41933353	0.009103993
Emotional Intelligence	1.00000000	-0.112720570
Letter Sets	-0.11272057	1.000000000
Figure Classification	-0.12546414	0.456398124
GPA	0.07426018	0.063219463
	Figure Classification	GPA
CN_Security	-0.05474368	0.0004787008
CN_Acceptance	-0.05117797	0.0575669887
CN_Initiative	-0.21344052	-0.0924685729
CN_Competence	-0.11371674	-0.0263324090
CN_Meaning	-0.16296513	-0.0727772843
M_Safety	-0.15193046	-0.0813071199
M_Belonging	-0.02276039	0.1023350503
M_Self-esteem	-0.09186538	-0.0292002029
M_Self-actualization	-0.13448426	-0.1789270226
EI_Challenge	0.03202353	0.1107931879
EI_Curiosity	-0.04457247	0.0725089358
EI_Independence Mastery	0.04283845	0.0994581347
EI_Easy work	-0.14602276	-0.1590620210
EI_Pleasing	-0.10623666	-0.1435770263
EI_Dependence on teacher	-0.15483213	0.0268236691
MC_Self-achievement	-0.02666903	0.0856796139
MC_Power	-0.11333337	0.0653286351
NE_Neuroticism	0.05266175	-0.0109055134
NE_Openness	0.08411218	0.0061354410
NE_Conscientiousness	-0.06315769	0.2057060914
Emotional Intelligence	-0.12546414	0.0742601785
Letter Sets	0.45639812	0.0632194628
Figure Classification	1.00000000	0.1114047545
GPA	0.11140475	1.0000000000

[illegible]

	CN_Cu- rarity	Sec- tand	Acc- tend	NI- pe- ing	Com- CN_M- Mong	M- Self- Safety	Self- actual- izing	CI- Mas- ion- ery	EI_In- de- pen- ence	EI_De- pen- ence	Elas- Plea- MC	Self- ti- MC	NE_Ne- ti- Open- ness	Emo- tel- gence	Figure Clas- si- fi- ca- tion GPA									
EI_Day- work	0.00	-	-	-	0.22	0.05	0.03	-	-	-	1.00	0.64	0.29	-	0.06	0.31	-	-	-	-	-	-		
EI_Plea- ing	0.01	-	0.00	-	0.15	0.05	0.04	-	-	-	0.64	1.00	0.25	-	0.02	0.21	-	-	-	0.01	-	-		
EI_De- pen- ence on teacher	0.18	0.08	0.11	0.17	0.19	0.22	0.16	0.09	-	0.06	-	0.29	0.25	1.00	-	0.16	0.07	-	0.14	0.07	-	-	0.03	
MC_S270- achievement	0.20	0.23	0.42	0.45	0.38	0.19	0.38	0.48	0.45	0.59	0.50	0.38	-	-	-	1.00	0.16	-	0.21	0.45	0.50	0.08	-	0.09
MC_Plea- ing	0.08	0.19	0.24	0.16	0.26	0.19	0.30	0.18	0.07	0.13	0.03	0.06	0.02	0.16	0.16	1.00	0.04	0.01	0.00	0.18	-	-	0.07	
NE_Ne- roti- cism	0.10	-	0.01	0.00	0.18	0.02	0.00	-	-	-	-	0.31	0.21	0.07	-	0.04	1.00	0.04	-	-	0.03	0.05	-	
NE_Open- ness	0.18	0.11	0.05	0.08	-	0.11	0.10	0.26	0.22	0.24	0.06	-	-	-	0.21	0.01	0.04	1.00	-	0.25	0.06	0.08	0.01	
NE_C230- sci- en- tious- ness	0.20	0.20	0.25	0.28	0.30	0.24	0.30	0.28	0.19	0.33	0.30	0.26	-	-	0.14	0.45	0.00	-	1.00	0.42	0.01	-	0.21	
Emo- tional	0.29	0.28	0.43	0.38	0.39	0.15	0.41	0.46	0.50	0.44	0.51	0.26	-	-	0.07	0.50	0.18	-	0.25	0.42	1.00	-	-	0.07
In- tel- li- gence	-	-	-	-	-	0.03	-	-	0.02	-	0.06	-	0.01	-	0.08	-	0.03	0.06	0.01	-	1.00	0.46	0.06	
Let- ter Sets	0.06	0.08	0.12	0.06	0.17	0.12	0.03	0.12	0.11	0.07	0.12	0.10	-	-	-	-	-	-	0.11	-	-	-	-	

EI_Independence Mastery	2.422055e-02	5.228074e-02	1.394700e-07	1.408677e-06
EI_Easy work	4.912668e-02	9.771317e-01	4.107111e-02	4.060984e-01
EI_Pleasing	1.571764e-01	7.913709e-01	2.455348e-02	9.881464e-01
EI_Dependence on teacher	8.541507e-05	2.607401e-04	1.123173e-01	3.295376e-02
MC_Self-achievement	1.071607e-07	3.294393e-06	5.475254e-18	4.006327e-21
MC_Power	3.194918e-02	1.329737e-01	1.745514e-04	1.250407e-06
NE_Neuroticism	2.841773e-02	4.965648e-02	4.551030e-01	9.126406e-01
NE_Openness	2.952851e-02	4.847273e-04	3.685905e-02	3.286108e-01
NE_Conscientiousness	4.219358e-06	5.334895e-05	7.637040e-07	1.195996e-08
Emotional Intelligence	7.159933e-09	1.528227e-08	2.546930e-19	8.514770e-15
Letter Sets	2.366982e-01	1.016089e-01	1.448901e-02	2.275822e-01
Figure Classification	2.814591e-01	3.140314e-01	2.184218e-05	2.490281e-02
GPA	9.928392e-01	2.800613e-01	8.232427e-02	6.214653e-01
	CN_Meaning	M_Safety	M_Belonging	M_Self-esteem
CN_Security	4.218226e-24	4.195161e-27	2.297490e-34	1.901200e-38
CN_Acceptance	1.441885e-22	7.801063e-11	1.098864e-20	5.865710e-22
CN_Initiative	9.530217e-42	3.528039e-13	2.136247e-13	1.224476e-37
CN_Compotence	3.764424e-39	3.582455e-20	9.185018e-18	1.654883e-63
CN_Meaning	0.000000e+00	6.120313e-18	1.848991e-17	1.235836e-36
M_Safety	6.120313e-18	0.000000e+00	2.441343e-16	2.883005e-29
M_Belonging	1.848991e-17	2.441343e-16	0.000000e+00	8.702128e-42
M_Self-esteem	1.235836e-36	2.883005e-29	8.702128e-42	0.000000e+00
M_Self-actualization	3.620241e-24	6.369320e-10	2.427985e-18	1.768963e-42
EI_Challenge	6.756904e-11	5.100003e-01	3.990606e-03	4.343145e-06
EI_Curiosity	7.743251e-18	2.151843e-03	4.705416e-07	6.597613e-10
EI_Independence Mastery	4.031654e-06	2.452456e-03	1.231302e-01	9.065990e-04
EI_Easy work	5.648146e-01	1.578655e-05	3.563259e-01	5.149005e-01
EI_Pleasing	3.973544e-01	2.423623e-03	3.009919e-01	3.801722e-01
EI_Dependence on teacher	1.040360e-03	2.377202e-04	1.537369e-05	1.254413e-03
MC_Self-achievement	1.657400e-14	1.268547e-04	3.584985e-15	7.459690e-24

MC_Power	1.197959e-03	1.780613e-07	2.100144e-04	9.742908e-10
NE_Neuroticism	9.513371e-01	2.993283e-04	7.323249e-01	9.593922e-01
NE_Openness	1.240948e-01	1.539554e-02	2.857882e-02	4.135890e-02
NE_Conscientiousness	2.472655e-09	1.025002e-06	9.733330e-10	1.760392e-08
Emotional Intelligence	1.479268e-15	3.047959e-03	1.670912e-17	1.999909e-21
Letter Sets	6.395089e-04	1.588929e-02	5.482358e-01	5.452426e-01
Figure Classification	1.257884e-03	2.661463e-03	6.545008e-01	7.031651e-02
GPA	1.718555e-01	1.267852e-01	5.439651e-02	5.839857e-01
M_Self-actualization	8.721673e-15	1.141906e-01	1.096078e-05	
CN_Security	1.283307e-18	2.790957e-03	1.457471e-07	
CN_Acceptance	1.161368e-37	7.762927e-13	1.123117e-16	
CN_Initiative	1.833998e-24	6.653344e-08	8.597662e-11	
CN_Compotence	3.620241e-24	6.756904e-11	7.743251e-18	
CN_Meaning	6.369320e-10	5.100003e-01	2.151843e-03	
M_Safety	2.427985e-18	3.990606e-03	4.705416e-07	
M_Belonging	1.768963e-42	4.343145e-06	6.597613e-10	
M_Self-esteem	0.000000e+00	4.032643e-08	1.252383e-13	
M_Self-actualization	4.032643e-08	0.000000e+00	2.960615e-74	
EI_Challenge	1.252383e-13	2.960615e-74	0.000000e+00	
EI_Curiosity	8.395448e-03	2.287944e-38	3.813944e-23	
EI_Independence Mastery	2.580413e-01	9.708070e-57	6.545213e-27	
EI_Easy work	5.034113e-01	3.824392e-21	4.953768e-19	
EI_Pleasing	7.926631e-02	4.135048e-03	2.635739e-01	
EI_Dependence on teacher	2.078819e-20	2.013467e-37	6.888135e-26	
MC_Self-achievement	2.621561e-04	1.833499e-01	7.604970e-03	
MC_Power	1.760425e-02	3.624507e-07	1.375687e-05	
NE_Neuroticism	1.703815e-07	1.454126e-05	2.374650e-06	
NE_Openness	2.068732e-04	4.532824e-11	1.091802e-09	
NE_Conscientiousness	4.662723e-26	4.142080e-20	2.283648e-27	
Emotional Intelligence	2.039370e-02	7.053846e-01	2.395913e-02	
Letter Sets	7.907739e-03	5.288701e-01	3.806425e-01	
Figure Classification	7.198197e-04	3.719590e-02	1.734458e-01	
GPA				
EI_Independence Mastery	2.422055e-02	4.912668e-02	1.571764e-01	
EI_Easy work	5.228074e-02	9.771317e-01	7.913709e-01	
EI_Pleasing	1.394700e-07	4.107111e-02	2.455348e-02	
CN_Security	1.408677e-06	4.060984e-01	9.881464e-01	
CN_Acceptance	4.031654e-06	5.648146e-01	3.973544e-01	
CN_Initiative	2.452456e-03	1.578655e-05	2.423623e-03	
CN_Compotence	1.231302e-01	3.563259e-01	3.009919e-01	
CN_Meaning	9.065990e-04	5.149005e-01	3.801722e-01	
M_Safety	8.395448e-03	2.580413e-01	5.034113e-01	
M_Belonging				
M_Self-esteem				
M_Self-actualization				

EI_Challenge	2.287944e-38	9.708070e-57	3.824392e-21
EI_Curiosity	3.813944e-23	6.545213e-27	4.953768e-19
EI_Independence Mastery	0.000000e+00	6.785440e-17	1.112635e-07
EI_Easy work	6.785440e-17	0.000000e+00	1.143845e-45
EI_Pleasing	1.112635e-07	1.143845e-45	0.000000e+00
EI_Dependence on teacher	6.376561e-10	4.817746e-09	7.442386e-07
MC_Self-achievement	8.168185e-15	1.740972e-18	7.285393e-06
MC_Power	5.660156e-01	2.379849e-01	6.804128e-01
NE_Neuroticism	5.141127e-03	3.514019e-10	1.967609e-05
NE_Openness	2.435960e-01	3.281201e-08	1.031548e-04
NE_Conscientiousness	1.608920e-07	1.522514e-06	3.023931e-03
Emotional Intelligence	3.074344e-07	1.223045e-07	2.099524e-03
Letter Sets	2.047791e-01	1.515845e-01	7.972643e-01
Figure Classification	3.994667e-01	3.898791e-03	3.621527e-02
GPA	6.157935e-02	2.688477e-03	6.812687e-03

	EI_Dependence on teacher	MC_Self-achievement
CN_Security	8.541507e-05	1.071607e-07
CN_Acceptance	2.607401e-04	3.294393e-06
CN_Initiative	1.123173e-01	5.475254e-18
CN_Compotence	3.295376e-02	4.006327e-21
CN_Meaning	1.040360e-03	1.657400e-14
M_Safety	2.377202e-04	1.268547e-04
M_Belonging	1.537369e-05	3.584985e-15
M_Self-esteem	1.254413e-03	7.459690e-24
M_Self-actualization	7.926631e-02	2.078819e-20
EI_Challenge	4.135048e-03	2.013467e-37
EI_Curiosity	2.635739e-01	6.888135e-26
EI_Independence Mastery	6.376561e-10	8.168185e-15
EI_Easy work	4.817746e-09	1.740972e-18
EI_Pleasing	7.442386e-07	7.285393e-06
EI_Dependence on teacher	0.000000e+00	1.717988e-01
MC_Self-achievement	1.717988e-01	0.000000e+00
MC_Power	2.075042e-03	1.335740e-03
NE_Neuroticism	1.841822e-01	2.262590e-03
NE_Openness	2.897154e-01	3.439260e-05
NE_Conscientiousness	5.591082e-03	1.867903e-20
Emotional Intelligence	1.618337e-01	1.786422e-26
Letter Sets	2.111196e-02	1.297819e-01
Figure Classification	2.195498e-03	6.000031e-01
GPA	6.149714e-01	1.075483e-01

	MC_Power	NE_Neuroticism	NE_Openness
CN_Security	3.194918e-02	2.841773e-02	2.952851e-02
CN_Acceptance	1.329737e-01	4.965648e-02	4.847273e-04

CN_Initiative	1.745514e-04	4.551030e-01	3.685905e-02
CN_Competence	1.250407e-06	9.126406e-01	3.286108e-01
CN_Meaning	1.197959e-03	9.513371e-01	1.240948e-01
M_Safety	1.780613e-07	2.993283e-04	1.539554e-02
M_Belonging	2.100144e-04	7.323249e-01	2.857882e-02
M_Self-esteem	9.742908e-10	9.593922e-01	4.135890e-02
M_Self-actualization	2.621561e-04	1.760425e-02	1.703815e-07
EI_Challenge	1.833499e-01	3.624507e-07	1.454126e-05
EI_Curiosity	7.604970e-03	1.375687e-05	2.374650e-06
EI_Independence Mastery	5.660156e-01	5.141127e-03	2.435960e-01
EI_Easy work	2.379849e-01	3.514019e-10	3.281201e-08
EI_Pleasing	6.804128e-01	1.967609e-05	1.031548e-04
EI_Dependence on teacher	2.075042e-03	1.841822e-01	2.897154e-01
MC_Self-achievement	1.335740e-03	2.262590e-03	3.439260e-05
MC_Power	0.000000e+00	3.856246e-01	8.672181e-01
NE_Neuroticism	3.856246e-01	0.000000e+00	4.072739e-01
NE_Openness	8.672181e-01	4.072739e-01	0.000000e+00
NE_Conscientiousness	9.477866e-01	7.599209e-11	4.339477e-02
Emotional Intelligence	4.570487e-04	1.449941e-18	7.236909e-07
Letter Sets	5.514209e-02	5.188529e-01	2.209340e-01
Figure Classification	2.539790e-02	3.001888e-01	9.761059e-02
GPA	2.201559e-01	8.379891e-01	9.084210e-01
NE_Conscientiousness Emotional Intelligence			
CN_Security	4.219358e-06		7.159933e-09
CN_Acceptance	5.334895e-05		1.528227e-08
CN_Initiative	7.637040e-07		2.546930e-19
CN_Competence	1.195996e-08		8.514770e-15
CN_Meaning	2.472655e-09		1.479268e-15
M_Safety	1.025002e-06		3.047959e-03
M_Belonging	9.733330e-10		1.670912e-17
M_Self-esteem	1.760392e-08		1.999909e-21
M_Self-actualization	2.068732e-04		4.662723e-26
EI_Challenge	4.532824e-11		4.142080e-20
EI_Curiosity	1.091802e-09		2.283648e-27
EI_Independence Mastery	1.608920e-07		3.074344e-07
EI_Easy work	1.522514e-06		1.223045e-07
EI_Pleasing	3.023931e-03		2.099524e-03
EI_Dependence on teacher	5.591082e-03		1.618337e-01
MC_Self-achievement	1.867903e-20		1.786422e-26
MC_Power	9.477866e-01		4.570487e-04
NE_Neuroticism	7.599209e-11		1.449941e-18
NE_Openness	4.339477e-02		7.236909e-07
NE_Conscientiousness	0.000000e+00		4.850514e-18

	CN_Ce- cularity	CN_Se- candiv	CN_Ac- tivetend	CN_Ni- pe- ing	Com- M_Mong	M_Be- long	Self- Safety	Self- reant	Global- izing	Mas- tery	EI_In- de- pen- ence	EI_De- pen- dence	NE_Ne- ti- Open	Emo- tel- ence	Fig- ure Clas- si- fi- ca- tion GPA						
CN_Ac- cep- tance	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.05	0.98	0.79	0.00	0.13	0.05	0.00	0.00	0.10	0.31	0.28	
CN_Ni- tia- tive	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.04	0.02	0.11	0.00	0.00	0.46	0.04	0.00	0.00	0.08	
CN_Com- pe- tence	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.41	0.99	0.03	0.00	0.00	0.91	0.33	0.00	0.23	0.62	
CN_Mas- ing	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.56	0.40	0.00	0.00	0.00	0.95	0.12	0.00	0.00	0.17	
M_Self- Be- long- ing	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.51	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.00	0.02	0.13	
M_Be- long- ing	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.12	0.36	0.30	0.00	0.00	0.00	0.73	0.03	0.00	0.55	0.65	0.05
M_Self- esteem	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.51	0.38	0.00	0.00	0.00	0.96	0.04	0.00	0.53	0.58	
M_Self- actualization	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.26	0.50	0.08	0.00	0.00	0.02	0.00	0.00	0.02	0.01	0.00
EI_Ou- lge	0.00	0.00	0.00	0.00	0.51	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.18	0.00	0.00	0.00	0.71	0.53	0.04
EI_Ou- rios- ity	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.26	0.00	0.00	0.00	0.00	0.00	0.02	0.38	0.17
EI_In- de- pen- dence	0.00	0.00	0.00	0.00	0.00	0.12	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.57	0.01	0.24	0.00	0.20	0.40	0.06
Mas- tery	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
EI_De- pendence	0.98	0.04	0.41	0.56	0.00	0.36	0.51	0.26	0.00	0.00	0.00	0.00	0.00	0.00	0.24	0.00	0.00	0.00	0.15	0.00	0.00
EI_Pleas- ure	0.79	0.02	0.99	0.40	0.00	0.30	0.38	0.50	0.00	0.00	0.00	0.00	0.00	0.00	0.68	0.00	0.00	0.00	0.80	0.04	0.01

	CN_Ce	CN_Ni	CN_Cu	CN_Pe	CN_Meng	CN_Meng	CN_Self	CN_Self	CN_Gen	CN_Mas	EI_In- de- pen- dence	EI_De- pen- dence	NE_Non- sci- tel-	Emo- tional	Fig- ure Clas- si- fi- ca- tion
EI_De- pen- dence on teacher	0.00	0.00	0.11	0.03	0.00	0.00	0.00	0.00	0.08	0.00	0.26	0.00	0.00	0.00	0.61
MC_Self achievement	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.17	0.00	0.00	0.11
MC_Power	0.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.18	0.10	0.57	0.24	0.68	0.00	0.22
NE_Non- roti- cism	0.05	0.46	0.91	0.95	0.00	0.73	0.96	0.02	0.00	0.00	0.01	0.00	0.18	0.00	0.84
NE_Open- ness	0.00	0.04	0.33	0.12	0.02	0.03	0.04	0.00	0.00	0.00	0.24	0.00	0.00	0.29	0.91
NE_Con- sci- en- tious- ness	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00
Emo- tional	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.16	0.00	0.16
In- tel- li- gence	0.24	0.10	0.10	0.23	0.00	0.02	0.55	0.55	0.02	0.71	0.02	0.20	0.15	0.80	0.24
Let- Sets	0.10	0.01	0.23	0.00	0.02	0.55	0.55	0.02	0.71	0.02	0.20	0.15	0.80	0.02	0.24
Fig- ure Clas- si- fi- ca- tion	0.28	0.31	0.00	0.02	0.00	0.00	0.65	0.07	0.01	0.53	0.38	0.40	0.00	0.04	0.04
GPA	0.99	0.28	0.08	0.62	0.17	0.13	0.05	0.58	0.00	0.04	0.17	0.06	0.00	0.10	0.00

```

r <- corr_test$r
p <- corr_test$p

stars <- ifelse(p < .001, "***", ifelse(p < .01, "**", ifelse(p < .05, "*", "")))

corr_table <- matrix(
  paste0(round(r, 2), stars),
  nrow = nrow(r),
  dimnames = dimnames(r))

corr_table |> kable(digits = 2)

```

	CN_Cu- rity	CN_Sec- ceptance	CN_Ni- tia-	CN_Com- pe-	M_Be- long- ing	Self- esteem	Self- efficacy	Chas- tity	Elas- ticity	MC_Pleas- ure	MC_Self- efficacy	NE_Net- tel-	Emo- tional sci- In-	Fig- ure Clas- si-
CN_Cu- rity	0.60***	0.53***	0.58***	0.48***	0.51***	0.57***	0.59***	0.38***	0.08	0.22***	0.11***	0.1*0.07	0.2***	0.29***
CN_Sec- ceptance	0.53***	0.55***	0.47***	0.32***	0.43***	0.46***	0.43***	0.15***	0.26***	0.1*0.0	0.18***	0.23***	0.08***	0.1***
CN_Ni- tia-	0.58***	0.55***	0.7***	0.61***	0.36***	0.36***	0.59***	0.35***	0.4***	0.26***	0.08	0.42***	0.19***	0.11***
CN_Com- pe-	0.48***	0.49***	0.71***	0.6***	0.44***	0.42***	0.79***	0.49***	0.27***	0.32***	0.24***	0.11***	0.45***	0.24***
M_Be- long- ing	0.51***	0.53***	0.46***	0.42***	0.4***	0.53***	0.31***	0.03	0.15***	0.15***	0.22***	0.19***	0.19***	0.26***
Self- esteem	0.57***	0.45***	0.36***	0.23***	0.41***	0.61***	0.42***	0.15***	0.25***	0.08	0.05	0.22***	0.38***	0.19***
Self- efficacy	0.59***	0.46***	0.59***	0.72***	0.58***	0.53***	0.61***	0.23***	0.31***	0.17***	0.03	0.04	0.16***	0.48***
Chas- tity	0.38***	0.43***	0.46***	0.43***	0.15***	0.26***	0.43***	0.15***	0.26***	0.1***	0.23***	0.08***	0.1***	0.2***
Elas- ticity	0.08	0.15***	0.18***	0.23***	0.08***	0.1***	0.23***	0.08***	0.1***	0.23***	0.08***	0.1***	0.23***	0.08***
MC_Pleas- ure	0.22***	0.11***	0.1***	0.07***	0.2***	0.11***	0.07***	0.2***	0.11***	0.07***	0.2***	0.11***	0.07***	0.2***
MC_Self- efficacy	0.11***	0.08***	0.07***	0.04***	0.15***	0.15***	0.22***	0.19***	0.26***	0.1***	0.23***	0.08***	0.1***	0.2***
NE_Net- tel-	0.45***	0.24***	0.19***	0.11***	0.42***	0.36***	0.59***	0.35***	0.4***	0.26***	0.08	0.42***	0.19***	0.11***
Emo- tional sci- In-	0.19***	0.11***	0.07***	0.04***	0.15***	0.15***	0.22***	0.19***	0.26***	0.1***	0.23***	0.08***	0.1***	0.2***
Fig- ure Clas- si-	0.29***	0.06	0.05	0.08	0.05	0.06	0.11	0.03	0.02	0.03	0.03	0.03	0.03	0.03

	Figure									
	EI_In-					Emo- Com- Clas-				
	de-					NE tel- Let-fi-				
	pen-					sci- In- si-				
	EI_Conce					NE_Non- tel- Let-fi-				
	CN_Sec-CN-CNi-Com-M_Be-					CN_Mem-M_Self-El-Cas-Mas-EI_Easy-Pleas-MC_Self-NE_tio-Opin-				
	cu-cep-tia-pe- CN_Mem-M_Self-El-Cas-Mas-EI_Easy-Pleas-MC_Self-NE_tio-Opin-					rityandvetendng M_isafesyeantudngitjionery workng teachachievMC_GrichRowssr-nessgencSetstion GPA				
M_Self-actualization	0.18***	0.43***	0.59***	0.49***	0.48***	0.31***	0.42***	0.62***	0.27***	0.36***
EI_Challenge	0.08***	0.15***	0.35***	0.27***	0.32***	0.03***	0.15***	0.23***	0.27***	0.76***
EI_Rios-ity	0.22***	0.26***	0.4***	0.32***	0.42***	0.15***	0.25***	0.31***	0.36***	0.76***
EI_De-pen-	0.11***	0.1***	0.26***	0.24***	0.25***	0.15***	0.08***	0.17***	0.13***	0.59***
Mas-tery	0.1***	0.1***	0.26***	0.24***	0.25***	0.15***	0.08***	0.17***	0.13***	0.59***
EI_Daily work	0.1***	0.1***	0.26***	0.24***	0.25***	0.15***	0.08***	0.17***	0.13***	0.59***
EI_Pleas-ing	0.11***	0.1***	0.26***	0.24***	0.25***	0.15***	0.08***	0.17***	0.13***	0.59***
EI_De-pen-	0.11***	0.1***	0.26***	0.24***	0.25***	0.15***	0.08***	0.17***	0.13***	0.59***
on teacher	0.11***	0.1***	0.26***	0.24***	0.25***	0.15***	0.08***	0.17***	0.13***	0.59***
MC_GrichRowssr-achievement	0.27***	0.23***	0.42***	0.45***	0.38***	0.19***	0.33***	0.45***	0.59***	0.38***
MC_Pleas-ing	0.18***	0.19***	0.24***	0.16***	0.26***	0.19***	0.33***	0.45***	0.59***	0.38***
NE_Non-roti-	0.11***	0.1***	0.26***	0.24***	0.25***	0.15***	0.08***	0.17***	0.13***	0.59***
ness	0.11***	0.1***	0.26***	0.24***	0.25***	0.15***	0.08***	0.17***	0.13***	0.59***

	CN_Ce	CN_Ac	CN_Ni	Com-M_Be	CN_Meng	M_Self	Self-Cor	CS_Mas	EI_In- de- pen- ence	EI_De- pen- ence	EL_Plea	MC_Self	6ti-NE	ti-NE	ti-NE	Emo- tel- genc	Fig- ure Clas- si- fi- ca- tion GPA	
NE_Ce	0.23**	0.25**	0.28**	0.30**	0.24**	0.28**	0.19**	0.33**	0.26***	-	0.14*	0.45**	-	-	1***	0.42**	0.01	0.21***
sci- en- tious- ness									0.24**	0.15**			0.32**	0.1*			0.06	
Emo- tional	0.29**	0.28**	0.33**	0.39**	0.15*	0.41**	0.46**	0.44**	0.26***	-	0.07	0.5**	0.18***	0.25**	0.42***	-	-	0.07
In- tel- li- gence									0.26**	0.16**			0.43***				0.11*	0.13*
Let- ter Sets	0.06	0.08	0.12*	0.06	0.17**	0.12*	0.03	0.12*	0.11*	0.07	0.12*	0.1			0.11*			
Fig- ure Clas- si- fi- ca- tion GPA	0.03	0.05	0.21*	0.11*	0.16*	0.15*	0.20	0.09	0.13**	0.04	0.15*	0.11*	0.03	0.11*		0.06	0.13*	
	0.06	-	-	-	0.1	-	-	0.11*	0.07	0.1	-	-	0.03	0.09	0.07	0.01	0.21**	0.07
		0.09	0.03	0.07	0.08		0.03	0.18***		0.16*	0.14**			0.01				0.11***

	CN_Security	CN_Acceptance	CN_Initiative	CN_Competence
CN_Security	1***	0.66***	0.53***	0.58***
CN_Acceptance	0.66***	1***	0.55***	0.49***
CN_Initiative	0.53***	0.55***	1***	0.7***
CN_Competence	0.58***	0.49***	0.7***	1***
CN_Meaning	0.48***	0.47***	0.61***	0.6***

M_Safety	0.51***	0.32***	0.36***	0.44***
M_Belonging	0.57***	0.45***	0.36***	0.42***
M_Self-esteem	0.59***	0.46***	0.59***	0.72***
M_Self-actualization	0.38***	0.43***	0.59***	0.49***
EI_Challenge	0.08	0.15**	0.35***	0.27***
EI_Curiosity	0.22***	0.26***	0.4***	0.32***
EI_Independence Mastery	0.11*	0.1	0.26***	0.24***
EI_Easy work	0.1*	0	-0.1*	-
0.04				
EI_Pleasing	0.07	0.01	-0.11*	0
EI_Dependence on teacher	0.2***	0.18***	0.08	0.11*
MC_Self-achievement	0.27***	0.23***	0.42***	0.45***
MC_Power	0.11*	0.08	0.19***	0.24***
NE_Neuroticism	0.11*	0.1*	-0.04	0.01
NE_Openness	0.11*	0.18***	0.11*	0.05
NE_Conscientiousness	0.23***	0.2***	0.25***	0.28***
Emotional Intelligence	0.29***	0.28***	0.43***	0.38***
Letter Sets	-0.06	-0.08	-0.12*	-
0.06				
Figure Classification	-0.05	-0.05	-0.21***	-
0.11*				
GPA	0	0.06	-0.09	-
0.03				
	CN_Meaning	M_Safety	M_Belonging	M_Self-esteem
CN_Security	0.48***	0.51***	0.57***	0.59***
CN_Acceptance	0.47***	0.32***	0.45***	0.46***
CN_Initiative	0.61***	0.36***	0.36***	0.59***
CN_Competence	0.6***	0.44***	0.42***	0.72***
CN_Meaning	1***	0.42***	0.41***	0.58***
M_Safety	0.42***	1***	0.4***	0.53***
M_Belonging	0.41***	0.4***	1***	0.61***
M_Self-esteem	0.58***	0.53***	0.61***	1***
M_Self-actualization	0.48***	0.31***	0.42***	0.62***
EI_Challenge	0.32***	0.03	0.15**	0.23***
EI_Curiosity	0.42***	0.15**	0.25***	0.31***
EI_Independence Mastery	0.23***	0.15**	0.08	0.17***
EI_Easy work	-0.03	0.22***	0.05	0.03
EI_Pleasing	-0.04	0.15**	0.05	0.04
EI_Dependence on teacher	0.17**	0.19***	0.22***	0.16**
MC_Self-achievement	0.38***	0.19***	0.38***	0.48***
MC_Power	0.16**	0.26***	0.19***	0.3***
NE_Neuroticism	0	0.18***	0.02	0
NE_Openness	0.08	-0.12*	0.11*	0.1*

NE_Conscientiousness	0.3***	0.24***	0.3***	0.28***
Emotional Intelligence	0.39***	0.15**	0.41***	0.46***
Letter Sets	-0.17***	-0.12*	0.03	-0.03
Figure Classification	-0.16**	-0.15**	-0.02	-0.09
GPA	-0.07	-0.08	0.1	-0.03
	M_Self-actualization	EI_Challenge	EI_Curiosity	
CN_Security		0.38***	0.08	0.22***
CN_Acceptance		0.43***	0.15**	0.26***
CN_Initiative		0.59***	0.35***	0.4***
CN_Competence		0.49***	0.27***	0.32***
CN_Meaning		0.48***	0.32***	0.42***
M_Safety		0.31***	0.03	0.15**
M_Belonging		0.42***	0.15**	0.25***
M_Self-esteem		0.62***	0.23***	0.31***
M_Self-actualization		1***	0.27***	0.36***
EI_Challenge		0.27***	1***	0.76***
EI_Curiosity		0.36***	0.76***	1***
EI_Independence Mastery		0.13**	0.59***	0.47***
EI_Easy work		-0.06	-0.69***	-0.51***
EI_Pleasing		-0.03	-0.45***	-0.43***
EI_Dependence on teacher		0.09	-0.14**	0.06
MC_Self-achievement		0.45***	0.59***	0.5***
MC_Power		0.18***	0.07	0.13**
NE_Neuroticism		-0.12*	-0.25***	-0.22***
NE_Openness		0.26***	0.22***	0.24***
NE_Conscientiousness		0.19***	0.33***	0.3***
Emotional Intelligence		0.5***	0.44***	0.51***
Letter Sets		-0.12*	0.02	-0.11*
Figure Classification		-0.13**	0.03	-0.04
GPA		-0.18***	0.11*	0.07
	EI_Independence Mastery	EI_Easy work	EI_Pleasing	
CN_Security		0.11*	0.1*	0.07
CN_Acceptance		0.1	0	0.01
CN_Initiative		0.26***	-0.1*	-0.11*
CN_Competence		0.24***	-0.04	0
CN_Meaning		0.23***	-0.03	-0.04
M_Safety		0.15**	0.22***	0.15**
M_Belonging		0.08	0.05	0.05
M_Self-esteem		0.17***	0.03	0.04
M_Self-actualization		0.13**	-0.06	-0.03
EI_Challenge		0.59***	-0.69***	-0.45***
EI_Curiosity		0.47***	-0.51***	-0.43***
EI_Independence Mastery		1***	-0.41***	-0.26***

EI_Easy work	-0.41***	1***	0.64***
EI_Pleasing	-0.26***	0.64***	1***
EI_Dependence on teacher	-0.31***	0.29***	0.25***
MC_Self-achievement	0.38***	-0.42***	-0.22***
MC_Power	0.03	0.06	0.02
NE_Neuroticism	-0.14**	0.31***	0.21***
NE_Openness	0.06	-0.28***	-0.2***
NE_Conscientiousness	0.26***	-0.24***	-0.15**
Emotional Intelligence	0.26***	-0.26***	-0.16**
Letter Sets	0.06	-0.07	0.01
Figure Classification	0.04	-0.15**	-0.11*
GPA	0.1	-0.16**	-0.14**
	EI_Dependence on teacher	MC_Self-achievement	MC_Power
CN_Security	0.2***	0.27***	0.11*
CN_Acceptance	0.18***	0.23***	0.08
CN_Initiative	0.08	0.42***	0.19***
CN_Competence	0.11*	0.45***	0.24***
CN_Meaning	0.17**	0.38***	0.16**
M_Safety	0.19***	0.19***	0.26***
M_Belonging	0.22***	0.38***	0.19***
M_Self-esteem	0.16**	0.48***	0.3***
M_Self-actualization	0.09	0.45***	0.18***
EI_Challenge	-0.14**	0.59***	0.07
EI_Curiosity	0.06	0.5***	0.13**
EI_Independence Mastery	-0.31***	0.38***	0.03
EI_Easy work	0.29***	-0.42***	0.06
EI_Pleasing	0.25***	-0.22***	0.02
EI_Dependence on teacher	1***	-0.07	0.16**
MC_Self-achievement	-0.07	1***	0.16**
MC_Power	0.16**	0.16**	1***
NE_Neuroticism	0.07	-0.15**	0.04
NE_Openness	-0.05	0.21***	0.01
NE_Conscientiousness	0.14**	0.45***	0
Emotional Intelligence	0.07	0.5***	0.18***
Letter Sets	-0.12*	0.08	-
0.1			
Figure Classification	-0.15**	-0.03	-
0.11*			
GPA	0.03	0.09	0.07
	NE_Neuroticism	NE_Openness	NE_Conscientiousness
CN_Security	0.11*	0.11*	0.23***
CN_Acceptance	0.1*	0.18***	0.2***
CN_Initiative	-0.04	0.11*	0.25***

CN_Competence	0.01	0.05	0.28***
CN_Meaning	0	0.08	0.3***
M_Safety	0.18***	-0.12*	0.24***
M_Belonging	0.02	0.11*	0.3***
M_Self-esteem	0	0.1*	0.28***
M_Self-actualization	-0.12*	0.26***	0.19***
EI_Challenge	-0.25***	0.22***	0.33***
EI_Curiosity	-0.22***	0.24***	0.3***
EI_Independence Mastery	-0.14**	0.06	0.26***
EI_Easy work	0.31***	-0.28***	-0.24***
EI_Pleasing	0.21***	-0.2***	-0.15**
EI_Dependence on teacher	0.07	-0.05	0.14**
MC_Self-achievement	-0.15**	0.21***	0.45***
MC_Power	0.04	0.01	0
NE_Neuroticism	1***	0.04	-0.32***
NE_Openness	0.04	1***	-0.1*
NE_Conscientiousness	-0.32***	-0.1*	1***
Emotional Intelligence	-0.43***	0.25***	0.42***
Letter Sets	0.03	0.06	0.01
Figure Classification	0.05	0.08	-0.06
GPA	-0.01	0.01	0.21***

Emotional Intelligence Letter Sets

CN_Security	0.29***	-0.06
CN_Acceptance	0.28***	-0.08
CN_Initiative	0.43***	-0.12*
CN_Competence	0.38***	-0.06
CN_Meaning	0.39***	-0.17***
M_Safety	0.15**	-0.12*
M_Belonging	0.41***	0.03
M_Self-esteem	0.46***	-0.03
M_Self-actualization	0.5***	-0.12*
EI_Challenge	0.44***	0.02
EI_Curiosity	0.51***	-0.11*
EI_Independence Mastery	0.26***	0.06
EI_Easy work	-0.26***	-0.07
EI_Pleasing	-0.16**	0.01
EI_Dependence on teacher	0.07	-0.12*
MC_Self-achievement	0.5***	0.08
MC_Power	0.18***	-0.1
NE_Neuroticism	-0.43***	0.03
NE_Openness	0.25***	0.06
NE_Conscientiousness	0.42***	0.01
Emotional Intelligence	1***	-0.11*

Letter Sets	-0.11*	1***
Figure Classification	-0.13*	0.46***
GPA	0.07	0.06
	Figure Classification	GPA
CN_Security	-0.05	0
CN_Acceptance	-0.05	0.06
CN_Initiative	-0.21***	-0.09
CN_Compotence	-0.11*	-0.03
CN_Meaning	-0.16**	-0.07
M_Safety	-0.15**	-0.08
M_Belonging	-0.02	0.1
M_Self-esteem	-0.09	-0.03
M_Self-actualization	-0.13**	-0.18***
EI_Challenge	0.03	0.11*
EI_Curiosity	-0.04	0.07
EI_Independence Mastery	0.04	0.1
EI_Easy work	-0.15**	-0.16**
EI_Pleasing	-0.11*	-0.14**
EI_Dependence on teacher	-0.15**	0.03
MC_Self-achievement	-0.03	0.09
MC_Power	-0.11*	0.07
NE_Neuroticism	0.05	-0.01
NE_Openness	0.08	0.01
NE_Conscientiousness	-0.06	0.21***
Emotional Intelligence	-0.13*	0.07
Letter Sets	0.46***	0.06
Figure Classification	1***	0.11*
GPA	0.11*	1***

```
# Open a large, high-resolution image
png(
  "correlation_plot.png",
  width = 16, height = 14, units = "in",
  res = 300, pointsize = 12, bg = "white"
)

corrplot::corrplot(
  round(corr_test$r, 1),
  method = "color",
  type = "lower",           # shows only half of the matrix
  diag = FALSE,            # removes the 1.00 diagonal
)
```

```

addCoef.col = "black",
number.cex = 0.85,          # makes correlation numbers smaller

number.font = 2,

tl.cex = 0.8,              # makes variable names smaller
tl.col = "black",
tl.srt = 45,

p.mat = corr_test$p,      # gives corrplot the p-values
sig.level = 0.05,
insig = "blank"           # hides nonsignificant correlations
)

# Finish saving the image
dev.off()

```

pdf
2

CFA

First Order

```

# Define CFA model

CN_model1 <- '

Security =~ CN_1_1 + CN_1_2 + CN_1_3 + CN_1_4 + CN_1_5 + CN_1_6 + CN_1_7 + CN_1_8

Acceptance =~ CN_2_1 + CN_2_2 + CN_2_3 + CN_2_4 + CN_2_5 + CN_2_6 + CN_2_7 + CN_2_8

Initiative =~ CN_3_1 + CN_3_2 + CN_3_3 + CN_3_4 + CN_3_5 + CN_3_6 + CN_3_7 + CN_3_8

Competence =~ CN_4_1 + CN_4_2 + CN_4_3 + CN_4_4 + CN_4_5 + CN_4_6 + CN_4_7 + CN_4_8

Meaning =~ CN_5_1 + CN_5_2 + CN_5_3 + CN_5_4 + CN_5_5 + CN_5_6 + CN_5_7 + CN_5_8

```

```

'

fit_CN1 <- cfa(CN_model1, data = data, std.lv = TRUE,
               missing = "fiml", estimator = "MLR")

summary(fit_CN1,
        fit.measures = TRUE,
        standardized = TRUE)

```

lavaan 0.7-2 ended normally after 127 iterations

Estimator	ML
Optimization method	NLMINB
Number of model parameters	130
Number of observations	390
Number of missing patterns	2

Model Test User Model:

	Standard	Scaled
Test Statistic	1687.560	1362.559
Degrees of freedom	730	730
P-value (Chi-square)	0.000	0.000
Scaling correction factor		1.239
Yuan-Bentler correction (Mplus variant)		

Model Test Baseline Model:

Test statistic	8926.287	7071.385
Degrees of freedom	780	780
P-value	0.000	0.000
Scaling correction factor		1.262

User Model versus Baseline Model:

Comparative Fit Index (CFI)	0.882	0.899
Tucker-Lewis Index (TLI)	0.874	0.893
Robust Comparative Fit Index (CFI)		0.903
Robust Tucker-Lewis Index (TLI)		0.896

Loglikelihood and Information Criteria:

Loglikelihood user model (H0)	-21998.892	-21998.892
Scaling correction factor for the MLR correction		1.219
Loglikelihood unrestricted model (H1)	-21155.112	-21155.112
Scaling correction factor for the MLR correction		1.236
Akaike (AIC)	44257.785	44257.785
Bayesian (BIC)	44773.384	44773.384
Sample-size adjusted Bayesian (SABIC)	44360.902	44360.902

Root Mean Square Error of Approximation:

RMSEA	0.058	0.047
90 Percent confidence interval - lower	0.054	0.044
90 Percent confidence interval - upper	0.062	0.051
P-value H ₀ : RMSEA ≤ 0.050	0.000	0.912
P-value H ₀ : RMSEA ≥ 0.080	0.000	0.000
Robust RMSEA		0.052
90 Percent confidence interval - lower		0.048
90 Percent confidence interval - upper		0.056
P-value H ₀ : Robust RMSEA ≤ 0.050		0.217
P-value H ₀ : Robust RMSEA ≥ 0.080		0.000

Standardized Root Mean Square Residual:

SRMR	0.055	0.055
------	-------	-------

Goodness of Fit Index:

Goodness of Fit Index (GFI)	0.891	
90 Percent confidence interval - lower	0.878	
90 Percent confidence interval - upper	0.903	
Robust GFI		0.910
90 Percent confidence interval - lower		0.896
90 Percent confidence interval - upper		0.923

Parameter Estimates:

Standard errors	Sandwich
Information bread	Observed
Observed information based on	Hessian

Latent Variables:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
Security =~						
CN_1_1	0.806	0.054	15.040	0.000	0.806	0.694
CN_1_2	0.896	0.065	13.678	0.000	0.896	0.678
CN_1_3	0.755	0.058	13.085	0.000	0.755	0.673
CN_1_4	0.769	0.076	10.057	0.000	0.769	0.505
CN_1_5	0.499	0.054	9.155	0.000	0.499	0.507
CN_1_6	0.586	0.063	9.252	0.000	0.586	0.551
CN_1_7	0.762	0.070	10.909	0.000	0.762	0.559
CN_1_8	0.833	0.060	13.969	0.000	0.833	0.685
Acceptance =~						
CN_2_1	1.007	0.063	15.946	0.000	1.007	0.690
CN_2_2	0.793	0.054	14.568	0.000	0.793	0.681
CN_2_3	0.992	0.061	16.158	0.000	0.992	0.738
CN_2_4	0.915	0.070	13.160	0.000	0.915	0.672
CN_2_5	0.950	0.058	16.287	0.000	0.950	0.742
CN_2_6	0.774	0.063	12.303	0.000	0.774	0.641
CN_2_7	0.951	0.065	14.635	0.000	0.951	0.713
CN_2_8	1.037	0.053	19.482	0.000	1.037	0.765
Initiative =~						
CN_3_1	0.815	0.063	12.848	0.000	0.815	0.617
CN_3_2	0.781	0.058	13.448	0.000	0.781	0.635
CN_3_3	0.707	0.049	14.366	0.000	0.707	0.655
CN_3_4	0.819	0.077	10.706	0.000	0.819	0.580
CN_3_5	0.852	0.071	12.056	0.000	0.852	0.618
CN_3_6	0.873	0.063	13.800	0.000	0.873	0.654
CN_3_7	0.848	0.062	13.583	0.000	0.848	0.672
CN_3_8	0.707	0.049	14.379	0.000	0.707	0.671
Competence =~						
CN_4_1	0.802	0.041	19.503	0.000	0.802	0.778
CN_4_2	0.770	0.039	19.791	0.000	0.770	0.765
CN_4_3	0.782	0.041	19.146	0.000	0.782	0.758
CN_4_4	0.776	0.043	17.899	0.000	0.776	0.717
CN_4_5	0.784	0.045	17.615	0.000	0.784	0.720
CN_4_6	0.773	0.049	15.657	0.000	0.773	0.705
CN_4_7	0.678	0.045	15.210	0.000	0.678	0.661
CN_4_8	0.668	0.051	13.049	0.000	0.668	0.634

Meaning =~

CN_5_1	1.172	0.064	18.325	0.000	1.172	0.811
CN_5_2	1.220	0.060	20.362	0.000	1.220	0.800
CN_5_3	1.209	0.059	20.322	0.000	1.209	0.790
CN_5_4	0.961	0.063	15.309	0.000	0.961	0.732
CN_5_5	1.102	0.066	16.685	0.000	1.102	0.713
CN_5_6	1.097	0.067	16.259	0.000	1.097	0.737
CN_5_7	1.045	0.064	16.224	0.000	1.045	0.776
CN_5_8	1.271	0.063	20.037	0.000	1.271	0.821

Covariances:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
Security ~~						
Acceptance	0.790	0.033	23.690	0.000	0.790	0.790
Initiative	0.648	0.048	13.420	0.000	0.648	0.648
Competence	0.670	0.039	17.259	0.000	0.670	0.670
Meaning	0.542	0.047	11.490	0.000	0.542	0.542
Acceptance ~~						
Initiative	0.636	0.049	13.055	0.000	0.636	0.636
Competence	0.545	0.046	11.843	0.000	0.545	0.545
Meaning	0.511	0.046	11.162	0.000	0.511	0.511
Initiative ~~						
Competence	0.809	0.034	23.799	0.000	0.809	0.809
Meaning	0.693	0.040	17.451	0.000	0.693	0.693
Competence ~~						
Meaning	0.656	0.041	15.807	0.000	0.656	0.656

Intercepts:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
.CN_1_1	5.569	0.059	94.737	0.000	5.569	4.797
.CN_1_2	5.203	0.067	77.752	0.000	5.203	3.937
.CN_1_3	5.792	0.057	102.033	0.000	5.792	5.167
.CN_1_4	5.164	0.077	66.929	0.000	5.164	3.389
.CN_1_5	6.072	0.050	121.798	0.000	6.072	6.168
.CN_1_6	5.964	0.054	110.699	0.000	5.964	5.605
.CN_1_7	5.264	0.069	76.310	0.000	5.264	3.864
.CN_1_8	5.441	0.062	88.253	0.000	5.441	4.469
.CN_2_1	4.592	0.074	62.160	0.000	4.592	3.148
.CN_2_2	5.544	0.059	94.021	0.000	5.544	4.761
.CN_2_3	5.151	0.068	75.713	0.000	5.151	3.834
.CN_2_4	5.200	0.069	75.349	0.000	5.200	3.815
.CN_2_5	5.361	0.065	82.611	0.000	5.361	4.183
.CN_2_6	5.623	0.061	92.049	0.000	5.623	4.661

.CN_2_7	5.123	0.068	75.838	0.000	5.123	3.840
.CN_2_8	4.977	0.069	72.499	0.000	4.977	3.671
.CN_3_1	5.067	0.067	75.761	0.000	5.067	3.836
.CN_3_2	5.041	0.062	80.984	0.000	5.041	4.101
.CN_3_3	5.554	0.055	101.614	0.000	5.554	5.145
.CN_3_4	5.023	0.072	70.198	0.000	5.023	3.555
.CN_3_5	4.905	0.070	70.253	0.000	4.905	3.557
.CN_3_6	4.974	0.068	73.655	0.000	4.974	3.730
.CN_3_7	5.318	0.064	83.255	0.000	5.318	4.216
.CN_3_8	5.718	0.053	107.170	0.000	5.718	5.427
.CN_4_1	5.685	0.052	108.932	0.000	5.685	5.516
.CN_4_2	5.626	0.051	110.344	0.000	5.626	5.587
.CN_4_3	5.815	0.052	111.371	0.000	5.815	5.639
.CN_4_4	5.662	0.055	103.249	0.000	5.662	5.228
.CN_4_5	5.605	0.055	101.571	0.000	5.605	5.143
.CN_4_6	5.667	0.056	102.085	0.000	5.667	5.169
.CN_4_7	5.833	0.052	112.331	0.000	5.833	5.688
.CN_4_8	5.782	0.053	108.434	0.000	5.782	5.491
.CN_5_1	5.262	0.073	71.864	0.000	5.262	3.639
.CN_5_2	4.946	0.077	64.091	0.000	4.946	3.245
.CN_5_3	5.156	0.077	66.538	0.000	5.156	3.369
.CN_5_4	5.397	0.066	81.239	0.000	5.397	4.114
.CN_5_5	5.069	0.078	64.789	0.000	5.069	3.281
.CN_5_6	5.121	0.075	67.965	0.000	5.121	3.442
.CN_5_7	5.372	0.068	78.786	0.000	5.372	3.989
.CN_5_8	5.056	0.078	64.500	0.000	5.056	3.266

Variances:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
.CN_1_1	0.699	0.070	9.935	0.000	0.699	0.519
.CN_1_2	0.944	0.103	9.183	0.000	0.944	0.541
.CN_1_3	0.688	0.075	9.127	0.000	0.688	0.547
.CN_1_4	1.730	0.142	12.189	0.000	1.730	0.745
.CN_1_5	0.720	0.070	10.275	0.000	0.720	0.743
.CN_1_6	0.788	0.085	9.326	0.000	0.788	0.696
.CN_1_7	1.275	0.108	11.801	0.000	1.275	0.687
.CN_1_8	0.788	0.094	8.383	0.000	0.788	0.531
.CN_2_1	1.115	0.093	12.045	0.000	1.115	0.524
.CN_2_2	0.727	0.066	11.049	0.000	0.727	0.537
.CN_2_3	0.822	0.105	7.812	0.000	0.822	0.455
.CN_2_4	1.020	0.095	10.692	0.000	1.020	0.549
.CN_2_5	0.739	0.079	9.416	0.000	0.739	0.450
.CN_2_6	0.857	0.090	9.551	0.000	0.857	0.589

.CN_2_7	0.876	0.115	7.613	0.000	0.876	0.492
.CN_2_8	0.763	0.081	9.398	0.000	0.763	0.415
.CN_3_1	1.080	0.095	11.405	0.000	1.080	0.619
.CN_3_2	0.902	0.083	10.895	0.000	0.902	0.597
.CN_3_3	0.666	0.062	10.695	0.000	0.666	0.571
.CN_3_4	1.326	0.137	9.675	0.000	1.326	0.664
.CN_3_5	1.175	0.122	9.664	0.000	1.175	0.618
.CN_3_6	1.017	0.104	9.738	0.000	1.017	0.572
.CN_3_7	0.872	0.076	11.434	0.000	0.872	0.548
.CN_3_8	0.610	0.057	10.697	0.000	0.610	0.549
.CN_4_1	0.419	0.037	11.330	0.000	0.419	0.395
.CN_4_2	0.421	0.040	10.531	0.000	0.421	0.415
.CN_4_3	0.452	0.043	10.486	0.000	0.452	0.425
.CN_4_4	0.571	0.063	9.056	0.000	0.571	0.487
.CN_4_5	0.573	0.051	11.278	0.000	0.573	0.482
.CN_4_6	0.604	0.065	9.231	0.000	0.604	0.503
.CN_4_7	0.593	0.063	9.480	0.000	0.593	0.563
.CN_4_8	0.663	0.074	8.977	0.000	0.663	0.598
.CN_5_1	0.716	0.084	8.481	0.000	0.716	0.342
.CN_5_2	0.835	0.070	11.905	0.000	0.835	0.359
.CN_5_3	0.880	0.091	9.685	0.000	0.880	0.376
.CN_5_4	0.799	0.077	10.393	0.000	0.799	0.464
.CN_5_5	1.174	0.122	9.616	0.000	1.174	0.492
.CN_5_6	1.010	0.110	9.172	0.000	1.010	0.456
.CN_5_7	0.721	0.079	9.100	0.000	0.721	0.398
.CN_5_8	0.781	0.095	8.209	0.000	0.781	0.326
Security	1.000				1.000	1.000
Acceptance	1.000				1.000	1.000
Initiative	1.000				1.000	1.000
Competence	1.000				1.000	1.000
Meaning	1.000				1.000	1.000

```
#Main Fit Indice
```

```
fitMeasures(fit_CN1, c("cfi", "tli", "rmsea", "srmr"))
```

```

    cfi    tli rmsea  srmr
0.882 0.874 0.058 0.055

```

```
#Corelations among factors
```

```
lavInspect(fit_CN1, "cor.lv")
```

```
Secrty Accptn Inittv Cmpntnc Meanng
```

Security	1.000				
Acceptance	0.790	1.000			
Initiative	0.648	0.636	1.000		
Competence	0.670	0.545	0.809	1.000	
Meaning	0.542	0.511	0.693	0.656	1.000

```
#modification indices
modificationIndices(fit_CN1,
                    sort. = TRUE,
                    minimum.value = 10)
```

	lhs	op	rhs	mi	epc	sepc.lv	sepc.all	sepc.nox
891	CN_3_5	~~	CN_3_6	75.981	0.537	0.537	0.491	0.491
451	CN_1_5	~~	CN_1_6	47.207	0.279	0.279	0.371	0.371
345	CN_1_2	~~	CN_1_8	36.636	0.313	0.313	0.363	0.363
1071	CN_5_4	~~	CN_5_5	33.102	0.312	0.312	0.322	0.322
154	Security	==	CN_3_6	26.368	-0.449	-0.449	-0.337	-0.337
343	CN_1_2	~~	CN_1_6	26.343	-0.251	-0.251	-0.291	-0.291
380	CN_1_3	~~	CN_1_6	24.631	0.207	0.207	0.281	0.281
151	Security	==	CN_3_3	22.487	0.336	0.336	0.311	0.311
851	CN_3_3	~~	CN_3_5	22.445	-0.236	-0.236	-0.267	-0.267
550	CN_1_7	~~	CN_5_6	21.680	0.288	0.288	0.254	0.254
342	CN_1_2	~~	CN_1_5	20.775	-0.211	-0.211	-0.256	-0.256
179	Acceptance	==	CN_1_7	20.158	0.568	0.568	0.417	0.417
346	CN_1_2	~~	CN_2_1	19.759	0.256	0.256	0.250	0.250
214	Initiative	==	CN_2_2	18.031	0.291	0.291	0.249	0.249
177	Acceptance	==	CN_1_5	17.893	-0.396	-0.396	-0.402	-0.402
186	Acceptance	==	CN_3_6	17.725	-0.344	-0.344	-0.258	-0.258
487	CN_1_6	~~	CN_1_8	17.215	-0.186	-0.186	-0.236	-0.236
591	CN_2_1	~~	CN_2_8	16.100	0.220	0.220	0.239	0.239
275	Meaning	==	CN_1_7	15.608	0.308	0.308	0.226	0.226
378	CN_1_3	~~	CN_1_4	14.806	0.235	0.235	0.215	0.215
771	CN_2_7	~~	CN_4_7	14.750	0.153	0.153	0.212	0.212
488	CN_1_6	~~	CN_2_1	14.692	-0.196	-0.196	-0.209	-0.209
153	Security	==	CN_3_5	13.748	-0.343	-0.343	-0.249	-0.249
675	CN_2_4	~~	CN_2_5	13.283	0.184	0.184	0.212	0.212
416	CN_1_4	~~	CN_1_6	12.784	0.225	0.225	0.193	0.193
232	Initiative	==	CN_5_4	12.653	0.276	0.276	0.210	0.210
495	CN_1_6	~~	CN_2_8	12.440	-0.153	-0.153	-0.198	-0.198
736	CN_2_6	~~	CN_3_5	11.536	-0.185	-0.185	-0.185	-0.185
637	CN_2_2	~~	CN_4_8	11.344	-0.127	-0.127	-0.183	-0.183
940	CN_3_7	~~	CN_5_4	11.325	0.156	0.156	0.187	0.187

1040	CN_4_7	~~	CN_5_4	11.123	0.125	0.125	0.182	0.182
350	CN_1_2	~~	CN_2_5	10.902	-0.158	-0.158	-0.189	-0.189
379	CN_1_3	~~	CN_1_5	10.662	0.129	0.129	0.183	0.183
1053	CN_5_1	~~	CN_5_2	10.497	0.154	0.154	0.199	0.199
176	Acceptance	==	CN_1_4	10.059	-0.459	-0.459	-0.301	-0.301
278	Meaning	==	CN_2_2	10.024	0.181	0.181	0.156	0.156
211	Initiative	==	CN_1_7	10.001	0.299	0.299	0.219	0.219

```
#Plot

# Open a high-resolution PNG
png(
  "CN1_CFA.png",
  width = 14, height = 22, units = "in",
  res = 300, pointsize = 12, bg = "white"
)

semPlot::semPaths(
  fit_CN1,
  what = "paths",
  whatLabels = "std",           # Standardized estimates
  layout = "tree2",
  rotation = 2,                 # Factors left, items right

  nCharNodes = 0,               # Show full variable names
  nDigits = 2,
  sizeMan = 6,
  sizeMan2 = 2.5,
  sizeLat = 10,
  sizeLat2 = 5,
  edge.label.cex = 0.9,

  residuals = FALSE,
  intercepts = FALSE,
  exoCov = TRUE,                # Show factor correlations

  color = list(lat = "#D8E8F5", man = "white"),
  edge.color = "black",
  fade = FALSE,
  normalize = FALSE,
  mar = c(4, 8, 4, 4)
)
```

```
# Finish saving
dev.off()
```

pdf
2

Second Order

```
# Define CFA model

CN_model2 <- '

Security =~ CN_1_1 + CN_1_2 + CN_1_3 + CN_1_4 + CN_1_5 + CN_1_6 + CN_1_7 + CN_1_8
Acceptance =~ CN_2_1 + CN_2_2 + CN_2_3 + CN_2_4 + CN_2_5 + CN_2_6 + CN_2_7 + CN_2_8
Initiative =~ CN_3_1 + CN_3_2 + CN_3_3 + CN_3_4 + CN_3_5 + CN_3_6 + CN_3_7 + CN_3_8
Competence =~ CN_4_1 + CN_4_2 + CN_4_3 + CN_4_4 + CN_4_5 + CN_4_6 + CN_4_7 + CN_4_8
Meaning =~ CN_5_1 + CN_5_2 + CN_5_3 + CN_5_4 + CN_5_5 + CN_5_6 + CN_5_7 + CN_5_8
CoreNeed =~ Security + Acceptance + Initiative + Competence + Meaning
'

fit_CN2 <- cfa(CN_model2, data = data, std.lv = TRUE,
              missing = "fiml", estimator = "MLR")

summary(fit_CN2,
        fit.measures = TRUE,
        standardized = TRUE)
```

lavaan 0.7-2 ended normally after 145 iterations

Estimator	ML
Optimization method	NLMINB
Number of model parameters	125

Number of observations	390
Number of missing patterns	2

Model Test User Model:

	Standard	Scaled
Test Statistic	1769.113	1427.514
Degrees of freedom	735	735
P-value (Chi-square)	0.000	0.000
Scaling correction factor		1.239
Yuan-Bentler correction (Mplus variant)		

Model Test Baseline Model:

Test statistic	8926.287	7071.385
Degrees of freedom	780	780
P-value	0.000	0.000
Scaling correction factor		1.262

User Model versus Baseline Model:

Comparative Fit Index (CFI)	0.873	0.890
Tucker-Lewis Index (TLI)	0.865	0.883
Robust Comparative Fit Index (CFI)		0.893
Robust Tucker-Lewis Index (TLI)		0.887

Loglikelihood and Information Criteria:

Loglikelihood user model (H0)	-22039.669	-22039.669
Scaling correction factor		1.214
for the MLR correction		
Loglikelihood unrestricted model (H1)	-21155.112	-21155.112
Scaling correction factor		1.236
for the MLR correction		
Akaike (AIC)	44329.338	44329.338
Bayesian (BIC)	44825.106	44825.106
Sample-size adjusted Bayesian (SABIC)	44428.489	44428.489

Root Mean Square Error of Approximation:

RMSEA	0.060	0.049
-------	-------	-------

90 Percent confidence interval - lower	0.056	0.046
90 Percent confidence interval - upper	0.064	0.053
P-value H_0: RMSEA <= 0.050	0.000	0.654
P-value H_0: RMSEA >= 0.080	0.000	0.000

Robust RMSEA		0.054
90 Percent confidence interval - lower		0.050
90 Percent confidence interval - upper		0.059
P-value H_0: Robust RMSEA <= 0.050		0.046
P-value H_0: Robust RMSEA >= 0.080		0.000

Standardized Root Mean Square Residual:

SRMR	0.062	0.062
------	-------	-------

Goodness of Fit Index:

Goodness of Fit Index (GFI)	0.883	
90 Percent confidence interval - lower	0.870	
90 Percent confidence interval - upper	0.895	
Robust GFI		0.902
90 Percent confidence interval - lower		0.888
90 Percent confidence interval - upper		0.916

Parameter Estimates:

Standard errors	Sandwich
Information bread	Observed
Observed information based on	Hessian

Latent Variables:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
Security =~						
CN_1_1	0.487	0.045	10.940	0.000	0.802	0.691
CN_1_2	0.527	0.051	10.358	0.000	0.868	0.657
CN_1_3	0.460	0.053	8.747	0.000	0.757	0.675
CN_1_4	0.481	0.063	7.583	0.000	0.793	0.520
CN_1_5	0.323	0.041	7.794	0.000	0.532	0.540
CN_1_6	0.372	0.046	8.107	0.000	0.613	0.576
CN_1_7	0.456	0.046	9.901	0.000	0.751	0.552
CN_1_8	0.494	0.050	9.856	0.000	0.813	0.668
Acceptance =~						

CN_2_1	0.689	0.059	11.743	0.000	1.012	0.694
CN_2_2	0.545	0.045	12.043	0.000	0.801	0.688
CN_2_3	0.667	0.059	11.329	0.000	0.980	0.730
CN_2_4	0.618	0.060	10.231	0.000	0.908	0.667
CN_2_5	0.650	0.058	11.130	0.000	0.955	0.745
CN_2_6	0.526	0.048	10.914	0.000	0.772	0.640
CN_2_7	0.645	0.059	10.947	0.000	0.947	0.710
CN_2_8	0.708	0.051	13.784	0.000	1.040	0.767
Initiative =~						
CN_3_1	0.356	0.053	6.670	0.000	0.823	0.623
CN_3_2	0.339	0.052	6.501	0.000	0.783	0.637
CN_3_3	0.311	0.045	6.874	0.000	0.717	0.665
CN_3_4	0.356	0.051	6.998	0.000	0.821	0.581
CN_3_5	0.364	0.063	5.732	0.000	0.839	0.609
CN_3_6	0.370	0.061	6.053	0.000	0.854	0.640
CN_3_7	0.369	0.060	6.099	0.000	0.851	0.675
CN_3_8	0.306	0.047	6.541	0.000	0.705	0.669
Competence =~						
CN_4_1	0.412	0.044	9.280	0.000	0.799	0.776
CN_4_2	0.397	0.038	10.366	0.000	0.770	0.765
CN_4_3	0.403	0.044	9.154	0.000	0.782	0.758
CN_4_4	0.399	0.045	8.935	0.000	0.775	0.716
CN_4_5	0.404	0.045	8.954	0.000	0.783	0.719
CN_4_6	0.399	0.045	8.896	0.000	0.774	0.706
CN_4_7	0.351	0.040	8.793	0.000	0.682	0.665
CN_4_8	0.344	0.040	8.618	0.000	0.667	0.633
Meaning =~						
CN_5_1	0.787	0.059	13.268	0.000	1.174	0.812
CN_5_2	0.818	0.062	13.191	0.000	1.220	0.800
CN_5_3	0.810	0.057	14.222	0.000	1.207	0.789
CN_5_4	0.643	0.047	13.614	0.000	0.959	0.731
CN_5_5	0.740	0.051	14.523	0.000	1.103	0.714
CN_5_6	0.736	0.056	13.256	0.000	1.098	0.738
CN_5_7	0.701	0.061	11.547	0.000	1.045	0.776
CN_5_8	0.853	0.063	13.543	0.000	1.272	0.821
CoreNeed =~						
Security	1.308	0.167	7.815	0.000	0.794	0.794
Acceptance	1.077	0.127	8.454	0.000	0.733	0.733
Initiative	2.080	0.344	6.043	0.000	0.901	0.901
Competence	1.663	0.219	7.597	0.000	0.857	0.857
Meaning	1.106	0.108	10.241	0.000	0.742	0.742

Intercepts:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
.CN_1_1	5.569	0.059	94.737	0.000	5.569	4.797
.CN_1_2	5.203	0.067	77.752	0.000	5.203	3.937
.CN_1_3	5.792	0.057	102.033	0.000	5.792	5.167
.CN_1_4	5.164	0.077	66.929	0.000	5.164	3.389
.CN_1_5	6.072	0.050	121.798	0.000	6.072	6.167
.CN_1_6	5.964	0.054	110.700	0.000	5.964	5.605
.CN_1_7	5.264	0.069	76.310	0.000	5.264	3.864
.CN_1_8	5.441	0.062	88.253	0.000	5.441	4.469
.CN_2_1	4.592	0.074	62.160	0.000	4.592	3.148
.CN_2_2	5.544	0.059	94.021	0.000	5.544	4.761
.CN_2_3	5.151	0.068	75.713	0.000	5.151	3.834
.CN_2_4	5.200	0.069	75.349	0.000	5.200	3.815
.CN_2_5	5.361	0.065	82.615	0.000	5.361	4.183
.CN_2_6	5.623	0.061	92.049	0.000	5.623	4.661
.CN_2_7	5.123	0.068	75.838	0.000	5.123	3.840
.CN_2_8	4.977	0.069	72.498	0.000	4.977	3.671
.CN_3_1	5.067	0.067	75.761	0.000	5.067	3.836
.CN_3_2	5.041	0.062	80.984	0.000	5.041	4.101
.CN_3_3	5.554	0.055	101.614	0.000	5.554	5.145
.CN_3_4	5.023	0.072	70.198	0.000	5.023	3.555
.CN_3_5	4.905	0.070	70.253	0.000	4.905	3.557
.CN_3_6	4.974	0.068	73.655	0.000	4.974	3.730
.CN_3_7	5.318	0.064	83.255	0.000	5.318	4.216
.CN_3_8	5.718	0.053	107.170	0.000	5.718	5.427
.CN_4_1	5.685	0.052	108.932	0.000	5.685	5.516
.CN_4_2	5.626	0.051	110.344	0.000	5.626	5.587
.CN_4_3	5.815	0.052	111.371	0.000	5.815	5.639
.CN_4_4	5.662	0.055	103.249	0.000	5.662	5.228
.CN_4_5	5.605	0.055	101.571	0.000	5.605	5.143
.CN_4_6	5.667	0.056	102.085	0.000	5.667	5.169
.CN_4_7	5.833	0.052	112.332	0.000	5.833	5.688
.CN_4_8	5.782	0.053	108.434	0.000	5.782	5.491
.CN_5_1	5.262	0.073	71.864	0.000	5.262	3.639
.CN_5_2	4.946	0.077	64.091	0.000	4.946	3.245
.CN_5_3	5.156	0.077	66.538	0.000	5.156	3.369
.CN_5_4	5.397	0.066	81.239	0.000	5.397	4.114
.CN_5_5	5.069	0.078	64.789	0.000	5.069	3.281
.CN_5_6	5.121	0.075	67.965	0.000	5.121	3.442
.CN_5_7	5.372	0.068	78.786	0.000	5.372	3.989
.CN_5_8	5.056	0.078	64.500	0.000	5.056	3.266

Variances:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
.CN_1_1	0.705	0.071	9.891	0.000	0.705	0.523
.CN_1_2	0.993	0.106	9.329	0.000	0.993	0.569
.CN_1_3	0.685	0.078	8.823	0.000	0.685	0.545
.CN_1_4	1.693	0.140	12.083	0.000	1.693	0.729
.CN_1_5	0.686	0.070	9.833	0.000	0.686	0.708
.CN_1_6	0.756	0.088	8.637	0.000	0.756	0.668
.CN_1_7	1.291	0.111	11.608	0.000	1.291	0.696
.CN_1_8	0.821	0.099	8.279	0.000	0.821	0.554
.CN_2_1	1.105	0.093	11.847	0.000	1.105	0.519
.CN_2_2	0.714	0.065	10.911	0.000	0.714	0.527
.CN_2_3	0.844	0.110	7.673	0.000	0.844	0.468
.CN_2_4	1.032	0.096	10.767	0.000	1.032	0.556
.CN_2_5	0.730	0.077	9.440	0.000	0.730	0.444
.CN_2_6	0.859	0.091	9.430	0.000	0.859	0.590
.CN_2_7	0.882	0.118	7.464	0.000	0.882	0.496
.CN_2_8	0.757	0.082	9.262	0.000	0.757	0.412
.CN_3_1	1.067	0.093	11.443	0.000	1.067	0.612
.CN_3_2	0.898	0.083	10.820	0.000	0.898	0.594
.CN_3_3	0.650	0.061	10.604	0.000	0.650	0.558
.CN_3_4	1.323	0.139	9.541	0.000	1.323	0.662
.CN_3_5	1.197	0.122	9.830	0.000	1.197	0.630
.CN_3_6	1.050	0.106	9.946	0.000	1.050	0.590
.CN_3_7	0.867	0.076	11.333	0.000	0.867	0.545
.CN_3_8	0.613	0.057	10.774	0.000	0.613	0.552
.CN_4_1	0.423	0.037	11.360	0.000	0.423	0.398
.CN_4_2	0.421	0.039	10.690	0.000	0.421	0.415
.CN_4_3	0.452	0.044	10.352	0.000	0.452	0.425
.CN_4_4	0.572	0.066	8.724	0.000	0.572	0.488
.CN_4_5	0.574	0.051	11.242	0.000	0.574	0.483
.CN_4_6	0.603	0.065	9.235	0.000	0.603	0.501
.CN_4_7	0.587	0.062	9.412	0.000	0.587	0.558
.CN_4_8	0.664	0.074	8.997	0.000	0.664	0.599
.CN_5_1	0.712	0.083	8.550	0.000	0.712	0.341
.CN_5_2	0.835	0.070	11.899	0.000	0.835	0.360
.CN_5_3	0.884	0.091	9.686	0.000	0.884	0.378
.CN_5_4	0.801	0.077	10.446	0.000	0.801	0.465
.CN_5_5	1.171	0.121	9.646	0.000	1.171	0.491
.CN_5_6	1.009	0.110	9.152	0.000	1.009	0.456
.CN_5_7	0.722	0.079	9.124	0.000	0.722	0.398
.CN_5_8	0.780	0.095	8.172	0.000	0.780	0.325
.Security	1.000				0.369	0.369
.Acceptance	1.000				0.463	0.463

.Initiative	1.000	0.188	0.188
.Competence	1.000	0.266	0.266
.Meaning	1.000	0.450	0.450
CoreNeed	1.000	1.000	1.000

```
#Main Fit Indice
fitMeasures(fit_CN2, c("cfi", "tli", "rmsea", "srmr"))
```

```
cfi    tli  rmsea  srmr
0.873 0.865 0.060 0.062
```

```
#Corelations among factors
lavInspect(fit_CN2, "cor.lv")
```

	Secrty	Accptn	Inittv	Cmptnc	Meanng	CoreNd
Security	1.000					
Acceptance	0.582	1.000				
Initiative	0.716	0.660	1.000			
Competence	0.681	0.628	0.772	1.000		
Meaning	0.589	0.543	0.668	0.636	1.000	
CoreNeed	0.794	0.733	0.901	0.857	0.742	1.000

```
#modification indices
modificationIndices(fit_CN2,
  sort. = TRUE,
  minimum.value = 10)
```

	lhs	op	rhs	mi	epc	sepc.lv	sepc.all	sepc.nox
928	CN_3_5	~~	CN_3_6	80.795	0.565	0.565	0.504	0.504
1118	Security	~~	Acceptance	71.140	0.733	0.733	0.733	0.733
382	CN_1_2	~~	CN_1_8	46.759	0.371	0.371	0.410	0.410
488	CN_1_5	~~	CN_1_6	40.308	0.253	0.253	0.351	0.351
151	Security	=~	CN_3_6	33.649	-0.355	-0.585	-0.439	-0.439
1108	CN_5_4	~~	CN_5_5	33.131	0.312	0.312	0.322	0.322
380	CN_1_2	~~	CN_1_6	28.632	-0.267	-0.267	-0.308	-0.308
379	CN_1_2	~~	CN_1_5	25.121	-0.235	-0.235	-0.285	-0.285
1124	Acceptance	~~	Competence	23.265	-0.461	-0.461	-0.461	-0.461
888	CN_3_3	~~	CN_3_5	22.765	-0.238	-0.238	-0.270	-0.270
383	CN_1_2	~~	CN_2_1	22.462	0.279	0.279	0.266	0.266
587	CN_1_7	~~	CN_5_6	20.980	0.285	0.285	0.250	0.250

176	Acceptance	==	CN_1_7	20.791	0.261	0.383	0.281	0.281
417	CN_1_3	~~	CN_1_6	20.614	0.190	0.190	0.264	0.264
150	Security	==	CN_3_5	20.278	-0.291	-0.479	-0.348	-0.348
524	CN_1_6	~~	CN_1_8	20.227	-0.205	-0.205	-0.260	-0.260
148	Security	==	CN_3_3	19.364	0.214	0.353	0.327	0.327
1119	Security	~~ Initiative	19.163	-0.596	-0.596	-0.596	-0.596	-0.596
211	Initiative	==	CN_2_2	16.007	0.124	0.286	0.245	0.245
272	Meaning	==	CN_1_7	15.980	0.222	0.330	0.242	0.242
171	Acceptance	==	CN_1_2	15.741	0.206	0.303	0.229	0.229
628	CN_2_1	~~	CN_2_8	15.191	0.215	0.215	0.236	0.236
808	CN_2_7	~~	CN_4_7	14.904	0.154	0.154	0.214	0.214
183	Acceptance	==	CN_3_6	13.976	-0.218	-0.320	-0.240	-0.240
712	CN_2_4	~~	CN_2_5	13.959	0.191	0.191	0.219	0.219
1127	Initiative	~~ Competence	13.853	0.608	0.608	0.608	0.608	0.608
304	CoreNeed	==	CN_1_7	13.517	0.497	0.497	0.365	0.365
415	CN_1_3	~~	CN_1_4	12.778	0.220	0.220	0.204	0.204
229	Initiative	==	CN_5_4	12.550	0.114	0.264	0.201	0.201
525	CN_1_6	~~	CN_2_1	12.534	-0.178	-0.178	-0.195	-0.195
773	CN_2_6	~~	CN_3_5	12.507	-0.195	-0.195	-0.192	-0.192
318	CoreNeed	==	CN_3_5	12.392	-0.833	-0.833	-0.604	-0.604
307	CoreNeed	==	CN_2_2	12.199	0.288	0.288	0.247	0.247
205	Initiative	==	CN_1_4	12.032	-0.186	-0.429	-0.282	-0.282
490	CN_1_5	~~	CN_1_8	11.815	-0.148	-0.148	-0.197	-0.197
208	Initiative	==	CN_1_7	11.733	0.162	0.374	0.274	0.274
977	CN_3_7	~~	CN_5_4	11.437	0.157	0.157	0.188	0.188
1077	CN_4_7	~~	CN_5_4	11.313	0.126	0.126	0.184	0.184
161	Security	==	CN_4_8	11.195	0.145	0.238	0.226	0.226
532	CN_1_6	~~	CN_2_8	11.015	-0.142	-0.142	-0.188	-0.188
471	CN_1_4	~~	CN_3_8	10.553	-0.181	-0.181	-0.178	-0.178
316	CoreNeed	==	CN_3_3	10.516	0.586	0.586	0.543	0.543
959	CN_3_6	~~	CN_5_3	10.456	0.175	0.175	0.182	0.182
1090	CN_5_1	~~	CN_5_2	10.269	0.152	0.152	0.197	0.197

PCA

```
# Select variables
data_PCA1 <- data[, c(
  "CN_S", "CN_A", "CN_I", "CN_C", "CN_M",
  "M_S", "M_B", "M_Se", "M_Sa",
  "EI_Ch", "EI_Cu", "EI_I", "EI_E", "EI_P", "EI_D",
```

```
"MC_S", "MC_P",
"NE_N", "NE_O", "NE_C", "EQ",
"LS", "FC", "gpa"
)]
```

```
# Remove rows with missing values
data_PCA1 <- na.omit(data_PCA1)
```

```
# PCA
pca_result1 <- prcomp(data_PCA1, center = TRUE, scale. = TRUE)
```

```
# Eigenvalues
eigenvalues1 <- pca_result1$sdev^2
eigenvalues1
```

```
[1] 6.6323449 3.6016259 1.6333038 1.4126361 1.2384872 1.1425682 1.0391148
[8] 0.8257460 0.7610668 0.6612288 0.6409935 0.5548794 0.4933352 0.4566079
[15] 0.4375701 0.4080714 0.3515617 0.3369506 0.2746212 0.2670617 0.2532661
[22] 0.2329592 0.1975483 0.1464512
```

```
# Kaiser rule: number of components with eigenvalue > 1
kaiser_components1 <- which(eigenvalues1 > 1)
kaiser_components1
```

```
[1] 1 2 3 4 5 6 7
```

```
n_components1 <- length(kaiser_components1)
n_components1
```

```
[1] 7
```

```
# Percentage of variance explained
variance_explained1 <- eigenvalues1 / sum(eigenvalues1)

variance_explained1 * 100
```

```
[1] 27.6347704 15.0067746 6.8054326 5.8859837 5.1603634 4.7607008
[7] 4.3296452 3.4406085 3.1711116 2.7551199 2.6708061 2.3119976
[13] 2.0555635 1.9025331 1.8232088 1.7002973 1.4648403 1.4039609
[19] 1.1442548 1.1127570 1.0552753 0.9706634 0.8231178 0.6102132
```

```
# Cumulative percentage of variance explained
cumulative_variance1 <- cumsum(variance_explained1)

cumulative_variance1 * 100
```

```
[1] 27.63477 42.64155 49.44698 55.33296 60.49332 65.25403 69.58367
[8] 73.02428 76.19539 78.95051 81.62132 83.93331 85.98888 87.89141
[15] 89.71462 91.41492 92.87976 94.28372 95.42797 96.54073 97.59601
[22] 98.56667 99.38979 100.00000
```

```
# PCA with Varimax rotation
pca_varimax1 <- psych::principal(
  data_PCA1,
  nfactors = n_components1,
  rotate = "varimax"
)

# Rotated component loadings
print(pca_varimax1$loadings, cutoff = 0.30)
```

Loadings:

	RC1	RC2	RC6	RC3	RC5	RC4	RC7
CN_S	0.827						
CN_A	0.745						
CN_I	0.758						
CN_C	0.783						
CN_M	0.715						
M_S	0.642				-0.393		
M_B	0.674						
M_Se	0.807						0.305
M_Sa	0.644				0.306		
EI_Ch		0.845					
EI_Cu	0.322	0.697					
EI_I		0.714					

EI_E	-0.820						
EI_P	-0.728						
EI_D	-0.422			0.464			
MC_S	0.440	0.483	0.357				
MC_P					0.873		
NE_N		-0.807					
NE_O				0.810			
NE_C	0.319		0.551	-0.353	0.364		
EQ	0.422		0.621				
LS				0.860			
FC				0.748			
gpa					0.782		

	RC1	RC2	RC6	RC3	RC5	RC4	RC7
SS loadings	5.675	3.698	1.799	1.663	1.355	1.297	1.213
Proportion Var	0.236	0.154	0.075	0.069	0.056	0.054	0.051
Cumulative Var	0.236	0.391	0.466	0.535	0.591	0.645	0.696

```
# Scree plot
scree1 <- data.frame(
  PC = 1:length(eigenvalues1),
  Eigenvalue = eigenvalues1,
  Variance = variance_explained1,
  CumulativeVariance = cumulative_variancel1,
  Keep = ifelse(eigenvalues1 > 1, "Keep", "Drop")
)

ggplot(scree1, aes(x = PC)) +
  geom_line(aes(y = Eigenvalue)) +
  geom_point(aes(y = Eigenvalue, color = Keep), size = 3) +
  geom_hline(yintercept = 1, linetype = "dashed") +

  geom_text_repel(
    data = subset(scree1, Keep == "Keep"),
    aes(
      y = Eigenvalue,
      label = paste0(
        "PC", PC, "\n",
        round(CumulativeVariance * 100, 1), "%"
      )
    )
  ),
```

```

    nudge_y = 0.2,
    size = 4
) +

geom_line(
  aes(y = CumulativeVariance * max(eigenvalues1)),
  linetype = "dotted"
) +

geom_point(
  aes(y = CumulativeVariance * max(eigenvalues1)),
  size = 2
) +

scale_y_continuous(
  name = "Eigenvalue",
  sec.axis = sec_axis(
    ~ . / max(eigenvalues1),
    name = "Cumulative Variance Explained",
    labels = scales::percent
  )
) +

labs(
  title = "Scree Plot with Kaiser Rule & Cumulative Variance",
  x = "Principal Components",
  color = "Kaiser Rule"
) +

theme_minimal()

```

Scree Plot with Kaiser Rule & Cumulative Variance

